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The phase field technique for modeling multiphase materials

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Abstract

This paper reviews methods and applications of the phase field technique, one of the fastest growing areas in computational materials science. The phase field method is used as a theory and computational tool for predictions of the evolution of arbitrarily shaped morphologies and complex microstructures in materials. In this method, the interface between two phases (e.g. solid and liquid) is treated as a region of finite width having a gradual variation of different physical quantities, i.e. it is a diffuse interface model. An auxiliary variable, the phase field or order parameter $\phi(\vec{x})$, is introduced, which distinguishes one phase from the other. Interfaces are identified by the variation of the phase field. We begin with presenting the physical background of the phase field method and give a detailed thermodynamical derivation of the phase field equations. We demonstrate how equilibrium and non-equilibrium physical phenomena at the phase interface are incorporated into the phase field methods. Then we address in detail dendritic and directional solidification of pure and multicomponent alloys, effects of natural convection and forced flow, grain growth, nucleation, solid–solid phase transformation and highlight other applications of the phase field methods. In particular, we review the novel phase field crystal model, which combines atomistic length scales with diffusive time scales. We also discuss aspects of quantitative phase field modeling such as thin interface asymptotic analysis and coupling to thermodynamic databases. The phase field methods result in a set of partial differential equations, whose solutions require time-consuming large-scale computations and often limit the applicability of the method. Subsequently, we review numerical approaches to solve the phase field equations and present a finite difference discretization of the anisotropic Laplacian operator.

(Some figures in this article are in colour only in the electronic version)

This article was invited by Professor M Finnis.

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1. Introduction

The field of diffusion controlled phase transformations is of considerable scientific and industrial interest. The manufacturing process of almost every man-made object involves a phase transformation at some stage. In metals phase transformations lead to different patterns, which can be analyzed by means of microscopy after special preparation procedures (e.g. etching).

Compositional and structural inhomogeneities appearing during the processing of metals may consist of spatially distributed phases of different composition and crystal structures, grains with different orientations, domains of different structural variants and structural defects. Understanding how such microstructures form and, subsequently, the modeling and simulations thereof is crucial for the development of improved materials. The complexity of the formed structures and their mutual interactions is non-trivial. This paper reviews the various research efforts that have been undertaken in order to gain understanding of the pattern formation processes in solidification, solid–solid transformations and related phenomena with the particularly powerful method of phase fields.

The paper is organized as follows: after some historical background leading to the formulation of the phase field equations (section 2.1) and the physical motivation of this approach (section 2.2) a thermodynamically consistent derivation of the phase field model for a binary alloy system is given (section 2.3) and the implications of the anisotropy of the surface free energy are discussed (section 2.4). In section 3 the various applications of the phase field methods are reviewed. Particular attention is paid to the case of dendritic solidification (section 3.1) for pure substances, binary alloys and multicomponent systems as well as the influence of natural convection and shear flow on the one hand and noise fluctuations and nucleation on the other hand. Directional solidification processes and grain growth are reviewed in sections 3.2 and 3.3, respectively. Eutectic, monotectic and peritectic growth is covered in section 3.4 while the novel phase field crystal approach is presented in section 3.5. Some other fields where phase fields have been applied are summarized in section 3.6. The rescaling for a quantitative modeling and solid–solid transformations are finally treated in sections 3.7 and 3.8. Section 4 covers the algorithmic and numerical aspects of the implementation of the phase field method. In particular, finite difference methods and adaptive grid methods are reviewed and the computational performance of the numerical methods is evaluated. Section 4.5 finally shows a detailed finite difference discretization of the anisotropic 3D Laplacian operator. We conclude in section 5 where the achievements of the phase field methods are briefly summarized and outstanding problems and future directions are discussed.

2. The phase field method

2.1. Historical notes

The phase field method is based on the concept of a diffuse interface, which was first considered by van der Waals in the late 19th century. J W Gibbs treated the thermodynamics of diffuse interface transitions shortly thereafter. The origins of the phase field equations lie in the works of Cahn and Hilliard [1] on the free energy of nonuniform systems and Allen and Cahn [2] on antiphase boundary motion.

Pivotal theoretical developments on the phase field models occurred in the 1980s that led to the modern use of phase field methods in materials science. Early works investigated dendritic solidification of pure and binary materials. Langer [3] and Fix [4] were the first to introduce a phase field model for the first-order phase transition. A similar diffuse interface model for solidification was independently developed by Collins and Levin [5]. Langer and Sekerka [6] considered a model of diffuse interface motion in a binary alloy system with a miscibility gap in a solid solution phase. Significant early work on the development and analysis of the phase field models of solidification was carried out by Caginalp *et al* [7–11]. They investigated effects of the anisotropy and the convergence of the phase field equations to the conventional sharp interface models in the limit of vanishing phase interface thicknesses. Penrose and Fife [12, 13] provided a framework for deriving the phase field equations in a thermodynamically consistent way from a single entropy functional, which allowed for modeling of non-isothermal solidification.

Realistic simulations with the phase field method are concurrent with the development of fast computers. Kobayashi's pioneering simulations of the growth of two- and three-dimensional dendrites [14–16] in the early nineties demonstrated the potential of the phase field approach as a computational tool for modeling complicated microstructures. His work precipitated the rapid development of phase field calculations in materials science. Nowadays the phase field method is one of the most successful techniques for modeling microstructural evolution. In the following sections we will review theoretical and computational phase field studies which have been published over the past two decades. Reviews on phase field models can also be found in [17–19].

2.2. Physical motivation of the phase field model

The numerical modeling of microstructures, and generally of phase transformations, as well as the development of computationally efficient schemes is closely connected to the advances and availabilities of high speed computers and especially large amounts of random access memory capacities.

Generally two views on modeling materials can be adopted: (i) macroscopic and (ii) microscopic. The macroscopic description of a phase transformation, e.g. the evolution of a solidifying crystal structure in a pure undercooled melt, dates back to Stefan in the 19th century. The basic equations for the system are called the sharp interface model or the Stefan problem: with a non-dimensionalized temperature field $u = (T - T_\infty)/(L/c_p)$ with T_∞ the far-field temperature, L the latent heat of fusion and c_p the heat capacity, the diffusion equation is given as

$$\frac{\partial u}{\partial t} = D\nabla^2 u \tag{1}$$

in the solid and the liquid with the diffusion coefficient D_s and D_l, respectively (here $D = D_\mathrm{s} = D_\mathrm{l}$, symmetric description). Energy conservation is taken into account at the interface for the phase transformation

$$v_\mathrm{n} = D(\nabla u|_\mathrm{solid} - \nabla u|_\mathrm{liquid}) \cdot \vec{n}, \tag{2}$$

where $\vec{n}$ is the outward pointing normal vector of the interface and v_n is the normal interfacial velocity. Additionally the Gibbs–Thomson relation takes into account that the temperature at the interface deviates from the melting temperature T_m:

$$u|_\mathrm{interface} = \Delta - d\kappa - \beta v_\mathrm{n}, \tag{3}$$

with $\Delta = (T_\mathrm{m} - T_\infty)/(L/c_p)$ the non-dimensional undercooling. The parameter d denotes the capillary length, which is anisotropic for realistic materials and κ is the local curvature of the interface; hence, a curved interface possesses a temperature below the melting temperature and will thus grow even faster, leading to branched and competing structures. The kinetic coefficient β accounts for the fact that the temperature of a fast moving interface also decreases with respect to T_m. For the study of the microstructural evolution usually low undercoolings are chosen; therefore, the influence of the kinetic coefficient is neglected. However, for fast solidification rates the kinetic contribution βv_n becomes dominant.

In order to simulate the sharp interface model the domain is divided into solid regions, liquid regions and the interface. Thus, there is a step function between solid and liquid. To evolve the system, the propagation of the interface has to be calculated: it is therefore a moving boundary problem. Details on the Stefan problem can be found in [20, 21]. While equations (1)–(3) are rather simple mathematically, a numerical implementation is a non-trivial task. Due to the complexity involved, only very few research groups have attempted to simulate 2D or even 3D structures in such a way [22–25].

On the other hand, the microscopic simulations (i.e. molecular dynamics MD) give interesting insights into phase transitions from the atomistic point of view. In particular, it is possible to determine material parameters such as the anisotropy of the surface free energy [26–28] and kinetic coefficients [29], which are often difficult to measure experimentally. While no direct measure for the interfacial position exists, it can be extracted by calculating the short range order parameter or the density. The solid–liquid interface on an atomistic level is not sharp but changes over a range of several, typically 5–8, atomic distances (interface thickness).

The downside of MD simulations for solidification of microstructures is obvious: typical structures are in the order of micrometres to millimetres, whereas atoms are ∼1Å; thus, the number of atoms needed to simulate anything of interest exceeds the computational capabilities of today's computers. In particular, since the dynamics is calculated by pairwise interactions, only time scales of picoseconds to nanoseconds can be simulated, whereas typical solidification times are in the range of milliseconds up to hours.

Thus, the phase field model fills in the gap between the two descriptions by coarse graining MD results of a diffuse interface and combining them in a PDE formulation with the macroscopic equations of the Stefan problem in such a way that for the vanishing interface thickness sharp interface equations are recovered. As shown in the subsequent sections this allows for the efficient solution of mesoscopic phenomena such as dendritic growth, grain growth and many others. However, the coarse graining of the microscopic results is also limiting the equations. In section 3.5 a novel phase field crystal approach (PFC) is reviewed, which positions itself between MD and the phase field method, keeping the spatial resolution of atoms, while being able to simulate on diffusive time scales.

2.3. *Mathematics of phase field equations*

There are different approaches to phase field modeling; the choice depends on the considered phenomena (e.g. pure melt or melt with multiple alloying elements, two or several transforming phases, solidification or solid–solid phase transformation) and the scientific background of the researcher (while materials scientists prefer to introduce realistic thermodynamics into the model, mathematicians concentrate on deriving higher-order accuracy methods). Therefore, one can distinguish phase field methods based on thermodynamical treatment with a single scalar order parameter (the so-called variational approach), methods derived to reproduce the traditional sharp interface approach in sharp and thin interface limits (non-variational methods) and methods employing multiple phase field order parameters. For a comprehensive description on thermodynamical derivation of the phase field equations the reader is referred to [18]. A geometrical approach for producing the phase field equations backwards from the sharp-interface method is given in [30]. The derivation clarifies the relationship between the phase field variable and the geometry of the solidifying front (i.e. curvature, velocity, etc). The thin interface approach, which is nowadays considered to be the most accurate, is presented in [31–33]. The mathematical background of this model is complicated; some basic ideas are discussed in section 3.7. Derivations of the phase field methods for multiphase transformations in multicomponent systems are given in [19, 34].

2.3.1. Thermodynamically consistent derivation. Nowadays the methodology of a thermodynamically consistent derivation of phase field models is well established and has been employed in many papers (e.g. [10, 35–39]). The derivation of the evolution equations is based on the basic concepts of irreversible thermodynamics. Here we present a rather general derivation of a binary alloy solidification following the work of Bi and Sekerka [40].

We first postulate a general form of an entropy functional S over the system volume V

$$S = \int_V \left(s(\phi, X, e) - \frac{\varepsilon_\phi^2}{2}|\nabla\phi|^2 - \frac{\varepsilon_X^2}{2}|\nabla X|^2 - \frac{\varepsilon_e^2}{2}|\nabla e|^2 \right) \mathrm{d}V, \tag{4}$$

where the thermodynamic entropy density s is a function of the phase field variable ϕ, the concentration X of a solute B in solvent A and the internal energy density e. The phase field variable is equal to 0 and 1 in solid and liquid, respectively, and varies between these two values over the phase interface. The entropy functional contains the gradient energy terms associated with the formation of an interface. The parameters ε_X, ε_e and ε_ϕ are constants.

The internal energy and mole fraction are conserved quantities; their evolution is governed by the normal conservation laws:

$$\dot{X} + \nabla \cdot \boldsymbol{J}_X = 0, \tag{5}$$

$$\dot{e} + \nabla \cdot \boldsymbol{J}_e = 0. \tag{6}$$

Consistent with the second law of thermodynamics, we postulate that the local entropy production is a non-negative quantity. This can be achieved by adopting the following linear laws of irreversible thermodynamics for the diffusional and heat fluxes:

$$\boldsymbol{J}_e = M_{ee}\nabla\left(\frac{\delta s}{\delta e}\right) + M_{eX}\nabla\left(\frac{\delta s}{\delta X}\right), \tag{7}$$

$$\boldsymbol{J}_X = M_{XX}\nabla\left(\frac{\delta s}{\delta X}\right) + M_{Xe}\nabla\left(\frac{\delta s}{\delta e}\right), \tag{8}$$

where M_{XX} and M_{ee} are related to the interdiffusional mobility of B and A and the heat conduction, respectively. The second law requires M_{ee} and M_{XX} to be positive. The coefficients M_{eX} and M_{Xe} describe the cross effects between the heat flow and diffusion and are equal according to Onsager's reciprocal theorem. It should be emphasized that the gradients $\nabla(\delta s/\delta X)$ and $\nabla(\delta s/\delta e)$ are to be evaluated isothermally and under fixed composition, respectively. Since there are gradient terms of X and e in the entropy functional (4), the variational derivatives in equations (7) and (8) are given by

$$\frac{\delta s}{\delta e} = \frac{\partial s}{\partial e} + \varepsilon_e^2\nabla^2 e, \tag{9}$$

$$\frac{\delta s}{\delta X} = \frac{\partial s}{\partial X} + \varepsilon_X^2\nabla^2 X. \tag{10}$$

The phase field variable is a non-conserved quantity; therefore, to guarantee that the entropy increases, a special relationship is derived

$$\dot{\phi} = M_\phi\frac{\delta s}{\delta\phi} = M_\phi\left(\frac{\partial s}{\partial\phi} + \varepsilon_\phi^2\nabla\phi\right), \tag{11}$$

where M_ϕ is a positive mobility related to the kinetic coefficient. Equation (11) is often called the Allen–Cahn equation [2].

For the case of a regular solution, the energy density and the entropy density can be written in the form

$$e(T, X, \phi) = e_A(T, \phi)(1 - X) + e_B(T, \phi)X + \Omega(\phi)X(1 - X), \tag{12}$$

$$s\left(e(T, X, \phi), X, \phi\right) = s_A(T, \phi)(1 - X) + s_B(T, \phi)X - \frac{R}{V_\mathrm{m}}\left[X\ln X + (1 - X)\ln(1 - X)\right], \tag{13}$$

where e_A and e_B are the energy densities of the pure components, s_A and s_B are their entropy densities, $\Omega(\phi)$ is a thermodynamic constant associated with the enthalpy of mixing, T is the temperature, R is the ideal gas constant and V_m is the molar volume, taken as a constant. We postulate the energy density of a pure material according to [35]

$$e_A(T, \phi) = e_A^\mathrm{S}(T)(1 - p(\phi)) + e_A^\mathrm{L}(T)p(\phi), \tag{14}$$

where e_A^L and e_A^S are energy densities for the liquid and solid phases, respectively, and $p(\phi)$ is an interpolating function, which satisfies $p(0) = 0$ and $p(1) = 1$. The entropy density s_A can then be found by integrating $\mathrm{d}s_A = \mathrm{d}e_A/T$ over the temperature at constant ϕ

$$s_A(T, \phi) = (1 - p(\phi))\int_0^T c_A^\mathrm{S}(u)\frac{\mathrm{d}u}{u} + p(\phi)\int_0^T c_A^\mathrm{L}(u)\frac{\mathrm{d}u}{u} + K_A(\phi), \tag{15}$$

where $c_A^\mathrm{S}(T)$ and $c_A^\mathrm{L}(T)$ are the specific heats of the liquid and solid phases, respectively, and $K_A(\phi)$ is an integration constant obeying $K_A(0) = K_A(1)$ according to the third law of thermodynamics. The corresponding relations for the component B can be obtained by replacing the subscript A by B.

The Helmholtz free energy density is constructed using equations (12) and (13):

$$f(T, X, \phi) = e(T, X, \phi) - Ts\left(e(T, X, \phi), X, \phi\right) = f_A(T, \phi)(1-X) + f_B(T, \phi)X + \Omega(\phi)X(1-X) + \frac{RT}{V_\mathrm{m}}\left[X\ln X + (1 - X)\ln(1 - X)\right], \tag{16}$$

where

$$f_{A,B}(T, \phi) = e_{A,B}(T, \phi) - Ts_{A,B}(T, \phi) \tag{17}$$

are the free energy densities of pure A and B. Substituting (14) and (15) into (17) yields an explicit expression for f_A

$$f_A(T, \phi) = f_A^\mathrm{S}(T, \phi)(1-p(\phi)) + f_A^\mathrm{L}(T, \phi)p(\phi) - TK_A(\phi), \tag{18}$$

where

$$f_A^{\mathrm{L,S}}(T,\phi) = e_A^{\mathrm{L,S}}(T) - T\int_0^T c_A^{\mathrm{L,S}}(u)\frac{\mathrm{d}u}{u} \tag{19}$$

with a similar expression for f_B.

In order to apply the evolution equations (5), (6) and (11) we need to evaluate the driving forces for the phase transformation. Namely, we need to calculate $\nabla(\partial s/\partial e)_{X,\phi}$, $\nabla(\partial s/\partial X)_{e,\phi}$ and $(\partial s/\partial\phi)_{e,X}$ terms in equations (9)–(11). By first noting that

$$\mathrm{d}e = T\,\mathrm{d}s + \left(\frac{\partial e}{\partial X}\right)_{s,\phi}\mathrm{d}X + \left(\frac{\partial e}{\partial \phi}\right)_{s,X}\mathrm{d}\phi \tag{20}$$

we obtain

$$\nabla\left(\frac{\partial s}{\partial e}\right)_{X,\phi} = \nabla\frac{1}{T} = -\frac{1}{T^2}\nabla T. \tag{21}$$

Furthermore, we use

$$\mathrm{d}f = -s\,\mathrm{d}T + \left(\frac{\partial e}{\partial X}\right)_{s,\phi}\mathrm{d}X + \left(\frac{\partial e}{\partial \phi}\right)_{s,X}\mathrm{d}\phi \tag{22}$$

to recognize

$$\left(\frac{\partial s}{\partial X}\right)_{e,\phi} = -\frac{1}{T}\left(\frac{\partial e}{\partial X}\right)_{s,\phi} = -\frac{1}{T}\left(\frac{\partial f}{\partial X}\right)_{T,\phi}, \tag{23}$$

$$\left(\frac{\partial s}{\partial \phi}\right)_{e,X} = -\frac{1}{T}\left(\frac{\partial e}{\partial \phi}\right)_{s,X} = -\frac{1}{T}\left(\frac{\partial f}{\partial \phi}\right)_{T,X}. \tag{24}$$

In order to evaluate $\nabla(\partial s/\partial X)_{e,\phi}$ we first write

$$\nabla\left(\frac{\partial s}{\partial X}\right)_{e,\phi} = \frac{\partial}{\partial\phi}\left(\frac{\partial s}{\partial X}\right)_{e,\phi}\nabla\phi + \frac{\partial}{\partial X}\left(\frac{\partial s}{\partial X}\right)_{e,\phi}\nabla X, \tag{25}$$

next, we calculate the second derivatives

$$\begin{aligned}\frac{\partial}{\partial\phi}\left(\frac{\partial s}{\partial X}\right)_{e,\phi} &= -\frac{1}{T}\frac{\partial^2 f}{\partial\phi\partial X}\\ &= H_A(T,\phi) - H_B(T,\phi) - \frac{1}{T}\frac{\mathrm{d}\Omega(\phi)}{\mathrm{d}\phi}(1-2X),\end{aligned} \tag{26}$$

$$\begin{aligned}\frac{\partial}{\partial X}\left(\frac{\partial s}{\partial X}\right)_{e,\phi} &= -\frac{1}{T}\left(\frac{\partial^2 f}{\partial X^2}\right)_{T,\phi}\\ &= -\frac{R}{V_{\mathrm{m}}}\frac{1}{X(1-X)} + \frac{2}{T}\Omega(\phi),\end{aligned} \tag{27}$$

where the following notation was introduced:

$$H_{A,B}(T,\phi) = \frac{1}{T}\frac{\partial f_{A,B}(T,\phi)}{\partial\phi}. \tag{28}$$

Substituting equations (26) and (27) into (25) gives

$$\begin{aligned}\nabla\left(\frac{\partial s}{\partial X}\right)_{e,\phi} &= \left[H_A(T,\phi) - H_B(T,\phi)\right.\\ &\quad\left. - \frac{1}{T}\frac{\mathrm{d}\Omega(\phi)}{\mathrm{d}\phi}(1-2X)\right]\nabla\phi\\ &\quad + \left[-\frac{R}{V_{\mathrm{m}}}\frac{1}{X(1-X)} + \frac{2}{T}\Omega(\phi)\right]\nabla X.\end{aligned} \tag{29}$$

And finally, the term in equation (11) is given by

$$\begin{aligned}\left(\frac{\partial s}{\partial\phi}\right)_{e,X} &= -\frac{1}{T}\left(\frac{\partial f}{\partial\phi}\right)_{T,X}\\ &= -H_A(T,\phi)(1-X) - H_B(T,\phi)X\\ &\quad - \frac{1}{T}\frac{\mathrm{d}\Omega(\phi)}{\mathrm{d}\phi}X(1-X).\end{aligned} \tag{30}$$

The general phase field equations for a binary alloy solidification can be obtained by substituting expressions for the fluxes into equations (9)–(11). Here we present a simplified version under the commonly used assumptions $\varepsilon_e = \varepsilon_X = 0$ and $M_{eX} = M_{Xe} = 0$:

$$\begin{aligned}\dot\phi &= M_\phi\left[\varepsilon_\phi^2\nabla^2\phi - H_A(T,\phi)(1-X) - H_B(T,\phi)X\right.\\ &\quad\left. -\frac{1}{T}\frac{\mathrm{d}\Omega}{\mathrm{d}\phi}X(1-X)\right],\\ \dot X &= \nabla\cdot\left[M_{XX}\left(\frac{R}{V_{\mathrm{m}}}\frac{1}{X(1-X)} - \frac{2}{T}\Omega(\phi)\right)\nabla X\right]\\ &\quad + \nabla\cdot\left[M_{XX}(H_B(T,\phi) - H_A(T,\phi)\right.\\ &\quad\left. + \frac{1}{T}\frac{\mathrm{d}\Omega}{\mathrm{d}\phi}(1-2X))\nabla\phi\right],\\ \frac{\partial e}{\partial T}\dot T + \frac{\partial e}{\partial\phi}\dot\phi + \frac{\partial e}{\partial X}\dot X &= \nabla\cdot\left[\frac{M_{ee}}{T^2}\nabla T\right].\end{aligned} \tag{31}$$

2.3.2. Reduction to the isothermal model of Warren and Boettinger. From the derived model, several existing phase field models can be recovered under certain assumptions. For example, a model of a pure substance by Wang *et al* [35] can be obtained by leaving out the diffusion equation and setting $X = 0$ in the phase field and heat equations. In this section we recover Warren and Boettinger's model [38] for isothermal binary alloy solidification. They assumed an ideal behavior of the solution, i.e. $\Omega(\phi) = 0$, so that the phase field and diffusion equations in (31) become equivalent to equations (3.2) and (3.6) in [38]. In order to continue, we evaluate $H_{A,B}$ explicitly using expression (18) for the free energy

$$H_A = p'(\phi)\frac{f_A^{\mathrm{L}}(T) - f_A^{\mathrm{S}}(T)}{T} - K_A'(\phi). \tag{32}$$

Since we are interested in temperatures near the melting point T_{M}, we can avoid integration from absolute zero (as in equation (19)) by integrating $\mathrm{d}(f_A/T)/\mathrm{d}(1/T) = e_A$ to obtain

$$\frac{f_A^{\mathrm{L}}(T) - f_A^{\mathrm{S}}(T)}{T} = -\int_{T_{\mathrm{M}}}^T\frac{L_A(u)}{u^2}\,\mathrm{d}u, \tag{33}$$

where we have used $f_A^{\mathrm{L}}(T_{\mathrm{M}}) = f_A^{\mathrm{S}}(T_{\mathrm{M}})$ and defined $L_A(T) = e_A^{\mathrm{L}}(T) - e_A^{\mathrm{S}}(T)$. Thus, $L_A(T_{\mathrm{M}})$ is just the latent heat (per unit volume) of A. To proceed further, we need to specify the energies $e_A^{\mathrm{L}}(T)$ and $e_A^{\mathrm{S}}(T)$. For a Ni–Cu alloy considered

in [38] one may assume that the heat capacity for the pure, single-phase materials is constant and that $c_A^{\mathrm{L}} = c_A^{\mathrm{S}}$. Then

$$e_A^{\mathrm{L}}(T) = e_A^{\mathrm{L}}(T_{\mathrm{M}}) + c_A^{\mathrm{L}}(T - T_{\mathrm{M}}^A), \tag{34}$$

$$e_A^{\mathrm{S}}(T) = e_A^{\mathrm{S}}(T_{\mathrm{M}}) + c_A^{\mathrm{S}}(T - T_{\mathrm{M}}^A), \tag{35}$$

which gives $e_A^{\mathrm{L}}(T) - e_A^{\mathrm{S}}(T) = e_A^{\mathrm{L}}(T_{\mathrm{M}}) - e_A^{\mathrm{S}}(T_{\mathrm{M}}) = L_A$ and

$$H_A(T,\phi) = -p'(\phi)L_A \int_{T_{\mathrm{M}}}^{T} \frac{1}{u^2}\,\mathrm{d}u - K_A'(\phi)$$
$$= p'(\phi)L_A\left(\frac{1}{T} - \frac{1}{T_{\mathrm{M}}^A}\right) - K_A'(\phi). \tag{36}$$

Choosing the interpolation function $p(\phi) = \phi^3(6\phi^2 - 15\phi + 10)$ and $K_A(\phi)$ as a double-well potential $K_A(\phi) = -W_A g(\phi)$, $g(\phi) = \phi^2(1-\phi)^2$, where W_A is the energy hump between the free energies of the liquid and solid, we obtain

$$H_A = W_A g'(\phi) + 30g(\phi)L_A\left(\frac{1}{T} - \frac{1}{T_{\mathrm{M}}^A}\right) \tag{37}$$

and similarly for B

$$H_B = W_B g'(\phi) + 30g(\phi)L_B\left(\frac{1}{T} - \frac{1}{T_{\mathrm{M}}^B}\right). \tag{38}$$

In the above expressions, T_{M}^A and T_{M}^B are the melting temperatures of the pure A and B and we have accounted for the fact that $p'(\phi) = 30g(\phi)$. Equations (37) and (38) are exactly the same as equation (3.3) in [38]. Taking the parameter $M_{XX} = V_{\mathrm{m}} D(\phi)X(1-X)/R$, where $D(\phi)$ is the diffusion coefficient, we fully recover the Warren–Boettinger model [38] for the isothermal solidification of the Ni–Cu alloy.

2.4. Anisotropy

For many materials, including metals, the surface energy and the kinetic coefficient depend on the orientation of the phase boundary. Since the anisotropy has a crucial impact on the shape of microstructures it is necessary to modify the phase field equations derived in section 2.3. The most widely used approach to include the anisotropy is to assume that the parameter ε_ϕ in equation (4) depends on the orientation of the interface with respect to the frame of reference through the anisotropic surface energy:

$$\sigma = \sigma_0 \eta(\vec{n}), \tag{39}$$

where σ_0 is the maximum surface energy. Then the variational derivative in equation (11) takes the form

$$\frac{\delta S}{\delta \phi} = \frac{\partial s}{\partial \phi} + \nabla\cdot(\eta^2(\vec{n})\nabla\phi) + \frac{\partial}{\partial x}\left(|\nabla\phi|^2\eta(\vec{n})\frac{\partial\eta(\vec{n})}{\partial\phi_x'}\right)$$
$$+ \frac{\partial}{\partial y}\left(|\nabla\phi|^2\eta(\vec{n})\frac{\partial\eta(\vec{n})}{\partial\phi_y'}\right) + \frac{\partial}{\partial z}\left(|\nabla\phi|^2\eta(\vec{n})\frac{\partial\eta(\vec{n})}{\partial\phi_z'}\right). \tag{40}$$

The choice of the anisotropy function $\eta(\vec{n})$ strictly depends on the modeled microstructure. In the case of three-dimensional dendritic growth the common choice is [31]

$$\eta(\vec{n}) = (1-3\gamma)\left[1 + \frac{4\gamma}{1-3\gamma}\frac{(\phi_x')^4 + (\phi_y')^4 + (\phi_z')^4}{|\nabla\phi|^4}\right], \tag{41}$$

which reduces to the standard four-fold variation of η in two dimensions

$$\eta(\theta) = 1 + \gamma\cos(4\theta), \tag{42}$$

where γ is the strength of the anisotropy and $\theta = \arctan(\phi_y'/\phi_x')$ gives an approximation of the angle between the interface and the orientation of the lattice. With this method of including the anisotropy, an asymptotic analysis for the interface width approaching zero yields the same form of the anisotropic Gibbs–Thomson equation that is employed for sharp interface theories [41]. The anisotropic Gibbs–Thomson equation requires $\sigma + \sigma''$ to be positive, which results in the restriction on the value of $\gamma < 1/15$, implying that we may use only mild values of anisotropy for simulating dendrites. While simulating dendritic growth with very small anisotropies the implicit anisotropy caused by the computational grid may significantly alter the growth morphology [42]. A method allowing one to evaluate the effective anisotropy and to reduce the effect of the grid anisotropy was proposed by Karma and Rappel [31].

In other cases, such as faceted morphologies, highly anisotropic interfacial properties are required. When $\sigma + \sigma''$ changes sign, this gives rise to orientations that are forbidden and a crystal interface with missing orientations develops. The formation of flat sides or facets can be predicted by the Wulff construction [43], which determines the equilibrium shape of a crystal. Flat sides or facets grow if the polar plot of the anisotropy function shows a narrow minimum or cusp, respectively. Strong anisotropies of the kinetic coefficient [44] and the interfacial energy [45–49] were studied in the case of faceted solidification. An example of such an anisotropy function would be

$$\eta(\theta) = 1 + \gamma|\sin(\theta)|. \tag{43}$$

This function is non-differentiable at the cusps $\theta = n\pi$ and needs to be regularized [45]. In order to simulate the growth of Widmanstätten plates, very strong anisotropy of the surface energy (with γ up to 100) was used in [50]. A new approach of matched asymptotic expansion presented by Wheeler [51] allows one to consider the sharp edge energy from a diffuse description of an interface through the regularization of the underlying mathematical model.

3. Applications of the phase field method

3.1. Dendritic solidification and discussion of inherent separation of length scales

Probably the most fascinating and most studied microstructures in solidification are dendrites, which appear in metals and other materials, such as ceramics, polymers and even concrete. The name stems from the highly branched tree-like morphology: the Greek name for tree is $'\delta\epsilon\nu\delta\rho o\nu'$ (dendron). Figure 1(*a*) shows an example of an experimental xenon dendrite growing into its pure undercooled melt. Since xenon crystallizes in the fcc structure a four-fold symmetry for the fins of the dendrite is observed. At the ridges of these fins side branches grow. Figure 1(*b*) shows an equiaxed dendritic structure computed in three dimensions with the phase field method.

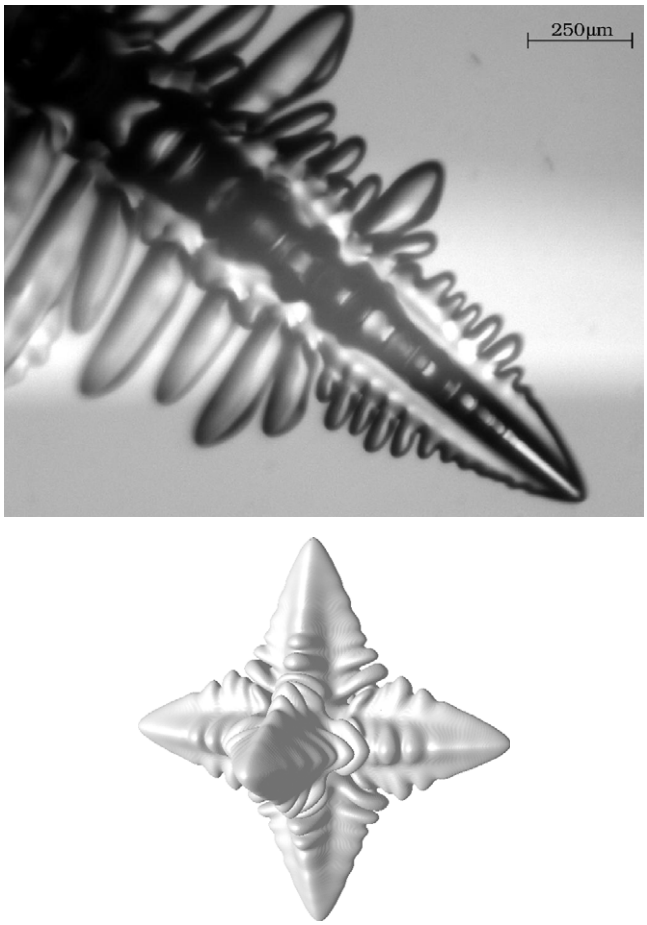


Figure 1. (*a*) A 3D experimental xenon dendrite growing into a pure undercooled melt by the authors [89]. Xenon crystallizes in the fcc structure; thus four fins develop. At the ridges of these fins side branches emerge. (*b*) A 3D equiaxed dendrite computed with a phase field model for binary alloy solidification calculated by the authors. The model assumes a cubic symmetry, which gives rise to six main dendritic arms. The secondary side branches appear due to thermal fluctuations in the melt.

In the computation cubic symmetry is assumed giving rise to six main dendritic arms.

Dendrites appear during solidification, which begins with the nucleation process. Once a nucleus has formed and starts growing the interface between the solid and liquid eventually becomes unstable due to the Mullins–Sekerka instability [52, 53] and breaks up. Given a high enough anisotropy of the crystalline lattice the structure develops into a dendritic morphology. After a transition period, the freely growing dendrite reaches a steady state where the dendrite tip advances with a constant tip velocity and a self-similar shape. The shape of the growing crystal is a result of an interplay between complex physical processes involving diffusion transport, capillarity and kinetics [54].

There are several control parameters that can be used to control the growth pattern, for example, the flow. In some industrial applications a homogeneous material is preferred, which means that dendritic structures are tried to be avoided, for example, by applying a forced flow to obtain a less heterogeneously solidified material. In other products, such as high quality turbine blades of super-alloys, the dendrites are forced to grow along the axis of the turbine blade, which results in improved thermal properties of the blades. The dendritic structure is probably the most common microstructure in metallurgy determining to a high degree the thermal and mechanical properties of solidified materials. Understanding the physics of dendritic growth is thus of great industrial and fundamental interest.

The first mathematical model of dendritic growth was presented by Ivantsov [55, 56]. He provided an analytical description of a paraboloid of revolution, representing the shape of an isothermal dendrite tip advancing with a steady-state velocity into a pure undercooled melt. In the absence of capillary and kinetic effects, the Ivantsov solution is formulated in terms of the Peclet number (product of the tip velocity V and the tip radius R). However, these quantities cannot be defined separately from the Ivantsov solution. The marginal stability hypothesis [52, 57] provides the second condition for selecting the tip operating point, namely, $R^2V = C/\sigma^\star$, where the constant C depends on the thermal diffusivity and capillary length and $\sigma^\star$ is the stability constant. According to microscopic solvability theory [58, 59], the stability constant depends on the strength of the surface energy anisotropy.

A mathematical description of solidification is given by the Stefan problem or the sharp interface model (see section 2.2). In the model, the evolution of diffusion fields is governed by partial differential equations and subject to boundary conditions at the solid–liquid interface. In spite of the apparent simplicity of the Stefan equations, analytical solutions are known only for a few special cases in a simple geometry, e.g. planar fronts. The phase field model avoids the problem of interface tracking of the sharp interface model, thus becoming very attractive for simulating pattern formation arising during phase transformations. Different aspects of the sharp interface and phase field models were recently reviewed by Sekerka [60].

Several length scales can be distinguished in the dendritic crystal. The smallest one is defined by the capillary length, which is of the order of a nanometre. The largest is set by the size of the microstructure and typically can be estimated as a millimetre. Another important length scale is the dendritic tip radius; its size depends on the solidifying material and growth conditions. The diffusion length can vary from nano- to millimetres, depending on the growth velocity. The diffusive nature of the phase field equations in principle allows for resolving all these length scales. However, this leads to the use of impractically large computational domains. Therefore it is crucial to develop phase field methods, which allow for a quantitative modeling even with unphysically thick diffuse interfaces.

3.1.1. Pure substance. A first computationally tractable phase field model derived in a thermodynamically consistent manner was presented by Wang *et al* [35] in 1993. By thermodynamical consistence one understands the derivation of the phase field equations from a single entropy (energy) functional in a way to guarantee local positive entropy production. A set of general conditions was developed in [35] to ensure that the phase field variable takes on constant

values in the bulk phases. Wheeler *et al* [36] performed an asymptotic analysis of the phase field model to recover the classical free-boundary problem in the limit of vanishing interface thicknesses and demonstrated how to relate phase field parameters to physical quantities. Nowadays the analysis of Wheeler [36] is often referred to in the literature as the sharp interface limit. Braun [61] performed a comprehensive comparison between the phase field model of [36] and the corresponding sharp interface model for the morphological stability of a planar interface.

The phase field model proposed in [36] significantly influenced the development of the phase field techniques and was extensively used, e.g. [37, 62–64], to study dendritic growth and to compare its different aspects with experimental observations. In particular, Murray *et al* [37] simulated the cleaving phenomenon, which involves the splitting of the dendrite tip into two branches and the subsequent predominant growth of one of them. Brush *et al* [65, 66] applied electrical current to a growing dendrite. The heat produced due to electrical current alters significantly the shape of a dendrite from the 4-fold symmetry. González-Cinca *et al* [67, 68] modified [36] by introducing an anisotropic diffusion coefficient to study the dynamics of a nematic–smectic-*B* interface. In that case, the angle between the primary dendritic arms differs from 90° and the overall envelope of the dendrite can be approximated as a rhomb. It was found that dendritic arms grow faster in the lower heat diffusion direction. Recently, Mullis [69] introduced a complex form of the anisotropy function into Wheeler's model [36] to allow for decoupling of capillary and kinetic anisotropies.

Despite the success of simulating realistic-looking dendrites, phase field calculations independent of numerical parameters could only be produced at high undercoolings as was demonstrated by Wang and Sekerka [63]. But even more importantly, in order to produce quantitative results, the phase interface thickness had to be set much smaller than the capillary length. However, using these thin interfaces would imply performing computations on grids larger than $10^6 \times 10^6 \times 10^6$ nodes (1 nm resolution). Even on today's computers utilizing these large computational domains is not feasible. Therefore, a new type of phase field model was developed in the mid-nineties.

Karma and Rappel [31, 32] proposed to revise the view on the thickness of the diffuse interface. They assumed that the interface thickness is small compared with the mesoscale of the diffusion field, but remains finite, and presented an analysis of the phase field equations in the thin interface limit. The derived phase field model [31, 32] produces accurate results with an interface thickness of the order of the capillary length, whereas in the previous models it had to be set much smaller than the capillary length. Moreover, parameters of the model could be adjusted to eliminate interface kinetics, thus allowing to perform simulations at low undercoolings (see section 3.7 for more details). Quantitative phase field simulations of free equiaxed dendritic growth have been carried out in three dimensions by Karma *et al* [31, 70, 71] at low undercoolings and by Bragard *et al* [72] at high undercoolings with the incorporation of anisotropic interface kinetic effects. For high undercoolings, the phase field simulations were performed using interfacial properties computed from atomistic molecular dynamics simulations [73].

The phase field model [32] was designed under the assumption of equal thermal conductivities in the liquid and solid phases. Almgren [74] demonstrated that in the case of unequal conductivities anomalous effects arise in the boundary conditions of the conventional sharp interface models. These effects include a temperature jump across the interface, an interface stretching and interface diffusion corrections to the Stefan condition. McFadden *et al* [75] applied thin interface asymptotics to derive a phase field model for pure solidification with unequal thermal conductivities. Their model is thermodynamically consistent and is based on independent entropy and internal energy functionals. Kim *et al* [76] proposed to suppress the anomalous interfacial effects by reducing the interface diffuseness. They decoupled the interface into two parts and localized the latent heat release (or solute redistribution) into a narrow region within the phase field interface. A systematic derivation of the sharp and thin interface limits of phase field models with general free energy functions was discussed by Elder *et al* [77].

Historically, dendritic growth is characterized by the tip radius and the velocity (the so-called operating point of a dendrite). It is based on two reasons: first, only for needle crystals (dendrite tip region) are analytical solutions known [55, 56]; second, the morphology of dendrites grown at different undercoolings is self-similar when rescaled with the respective tip radius. In many phase field studies, the computed results are validated by a comparison of the dendrite operating point with predictions of solvability theory [58]. Karma and Rappel [31, 70, 78] studied the operating point of a three-dimensional dendrite for a wide range of surface energy anisotropies. They found good agreement with the prediction of the solvability theory for small anisotropies, but poor agreement in the case of large anisotropies. This discrepancy is explained by the departure of the tip morphology from the axisymmetric approximation used in the solvability theory for large anisotropies. Effects of the ratio of solid to liquid conductivity on the operating point of a dendrite were studied by Mullis [79]. He showed that with the phase field model the ratio of thermal conductivities in the solid and liquid phases (thus also the stability parameter σ^*) changes more rapidly than that predicted by the solvability theory. Wang and Sekerka [64] numerically studied the dependence of the dendrite operating point on the anisotropy of surface tension and kinetics at large undercoolings. Jeong *et al* [80] presented an efficient and accurate method for extracting the dendrite tip position and the tip radius. Despite the overall agreement with solvability theory, a direct comparison of the phase field simulations with experimental observations is desirable, but not possible at the moment. Even with the well-developed phase field theory and modern computers, performing computations with the experimental undercoolings remains beyond computational capacity. To our knowledge, the lowest non-dimensional undercooling reported in phase field simulations is 0.05 [81], while the undercoolings in

experiments of, for example, dendritic growth of xenon is in the order of 0.001 [82].

For higher undercoolings kinetic effects start dominating over capillary effects. Vetsigian and Goldenfeld [83] proposed a phase field model which allows efficient computations in the regime of large interface kinetics. Mullis [84] demonstrated that at high undercoolings the relation $R^2V = \text{const}$ breaks; the tip radius as a function of the undercooling reaches a minimum. Further increase in the undercooling leads again to an increase in the tip radius. Therefore, two regions of dendritic growth can be identified: diffusion limited (low undercoolings) and kinetically limited (high undercoolings) growth.

Despite the success in describing dendritic tip growth, the analytical theories do not provide a mechanism for studying the development of the rest of the dendrite behind the tip. However, the dense sidebranching region of a dendrite is of industrial interest. Other parameters are needed to characterize the dendrite shape far from the tip. González-Cinca [85] argued that the use of integral parameters (as introduced for experimental structures in [86–88]), surface area and contour length, are more appropriate for the characterization of the dendritic shape of simulated structures in the kinetically dominated regime. Besides, the integral parameters can be directly evaluated in experiments [82, 88, 89]. So far little attention has been paid to the transient growth of dendrites, i.e. the period between nucleation and steady-state growth, even though this issue is of great practical importance and can be studied experimentally [82]. Steinbach *et al* [90] showed that the size of initial nuclei significantly influences the initial transient and subsequent growth of a dendrite.

Growth conditions determine the morphology of evolving crystals. Singer *et al* [91] studied the effects of the surface energy anisotropy and undercooling on growth morphologies, which include dendrites, seaweeds and doublons by means of two different phase field models. By performing a large number of simulations, a morphology phase diagram for solidification from pure melt in two and three dimensions was constructed. In agreement with experimental observations [82, 92] and analytical calculations [93, 94], seaweeds appear at low anisotropies, while dendrites grow at high anisotropies. In two dimensions, a doublon region was found at mild anisotropies and high undercoolings. However, this doublon region does not exist in three dimensions. Transient morphologies as found in experiments [82, 92], in particular doublons, in 2D and 3D, have been simulated by Singer *et al* [91, 95, 96]. It was found that the growing morphology is strongly dependent on the initial conditions: using a spherical seed in 3D, doublons were not observed. However, by using an ellipsoidal nucleus of the same volume, splitting to stable doublon growth was found with otherwise identical conditions. The formation of triplet structures in a channel was studied by Abel *et al* [97] for vanishing anisotropies. Singer *et al* [98] reported on the formation of triplet structures in experiments with xenon using a capillary shifting technique. Implementing at the same time the experimental procedure with a 3D phase field model they also found triplets for non-zero anisotropies [98].

Nestler *et al* [42] presented 3D morphology transitions for various anisotropies of the surface energy and kinetics at different undercoolings. A drastic influence of the kinetic anisotropy on the morphology of a solidifying crystal was demonstrated in [69]. Strong kinetic anisotropy alters the growth direction of primary branches and produces seaweed morphologies even at high values of the surface energy anisotropy. Mullis *et al* [99, 100] also argued that the transition from dendritic to seaweed morphology is responsible for spontaneous grain refinement in deeply undercooled metallic melts. Their simulations are consistent with experimental observations [101]. While at low (vanishing) anisotropies one expects seaweed-like structures, high values of anisotropy lead to faceted morphologies [44–47]. Other aspects, which can be attributed to the dendritic solidification of a pure melt, include incorporation of noise, effects of the anisotropy in the surface energy and kinetic coefficients as well as the influence of the fluid flow. These phenomena are discussed in the following sections.

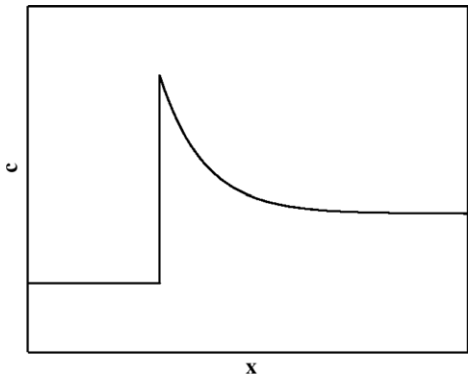


Figure 2. Sketch of the solute distribution at the solid–liquid interface in a binary alloy.

3.1.2. Binary alloys. The growth of a crystal from an alloy melt results in a local change in the composition over the phase boundary, since the equilibrium condition for a binary system requires the continuity of the chemical potentials of the components at the phase boundary. A typical concentration profile is shown in figure 2. The composition shows an almost constant value in the growing phase, then jumps across the interface and relaxes to its far-field value c_∞ in the parent phase. The difference in the composition at the moving interface, assuming local thermodynamic equilibrium under normal solidification conditions, can be described by the partition coefficient

$$k = \frac{c_S}{c_L}, \tag{44}$$

where c_S and c_L are the concentrations of the solute at the solid and liquid sides of the interface, respectively. The values of c_S and c_L for a given temperature can be obtained through the phase diagram. Under rapid solidification conditions k becomes a function of the interface velocity [102].

Wheeler *et al* [103, 104] and Boettinger *et al* [105] presented a phase field model for the solidification of a binary alloy. They included gradient energy contributions for the phase field and for the composition field. Wheeler *et al* [103, 104] performed an asymptotic analysis and related

the phase field parameters to the material and growth parameters in real systems. They also demonstrated Ostwald ripening (merging and coarsening of solid particles) in two dimensions and performed a rigorous study of the interface velocity dependence on the solute profiles by numerous 1D calculations. While at low solidification rates equilibrium behavior was recovered, at high solidification rates non-equilibrium effects such as solute trapping naturally emerged from the model. By solute trapping one understands the velocity dependence on the jump in concentration, which provides a mechanism whereby the jump vanishes at high rates of solidification leading to a partitionless transformation.

In subsequent works Ahmad *et al* [102] showed that solute trapping occurs when the solute diffusion length D_I/V is comparable to the diffuse interface thickness, where D_I is a characteristic solute diffusivity in the interfacial region. Kim *et al* [106] demonstrated that the finite interface thickness can result in the solute trapping effect, because partitioning of the solute atoms in the diffuse interface requires a finite relaxation time. Different aspects of solute trapping have been studied by Charach and Fife [107], Conti and Fermani [108] and Kim and Kim [109]. Loginova *et al* [50, 110] investigated the transition between diffusion controlled and massive (partitionless) transformation of austenite to ferrite phase transitions in Fe–C alloys. Since the simulations were performed in one dimension, resolving a physically realistic interface width of 1 nm did not represent a problem. Solute trapping was observed when the far-field C composition of austenite was below a critical value for a given temperature. In this case the solute profile comprises a spike traveling with constant velocity; the variation of the composition occurs inside the diffuse interface.

Warren and Boettinger [38] derived a phase field model for the isothermal solidification of a binary alloy, applying constant diffusivities within the solid and liquid phases, and performed 2D simulations of dendritic growth into a highly supersaturated liquid. This model has also been used to study directional solidification at high velocities [111], solidification during recalescence [112], Ostwald ripening and coalescence [113]. An asymptotic sharp interface analysis of the model was performed by Kessler [114]. George and Warren [115] presented impressive simulations of a 3D dendrite having a large number of secondary and ternary side branches (see figure 3). A desirable extension of the isothermal model is to study the effect of heat flow due to the release of latent heat. Inclusion of thermal diffusion into the model represents a severe numerical problem since the temperature and solute evolution occurs on completely different time- and length-scales. A simplified approach was proposed in [112], where the spatial variation of the temperature was neglected and the heat equation was replaced by a heat balance of an imposed heat extraction rate and the latent heat release rate.

Bi and Sekerka [40] proposed a general phase field model for binary alloy solidification, which included energy gradients of phase field, concentration and internal energy. Loginova *et al* [39] derived a model for simultaneous heat and solute evolution. The inherent difficulty of different time scales was circumvented by applying adaptive grids and implicit

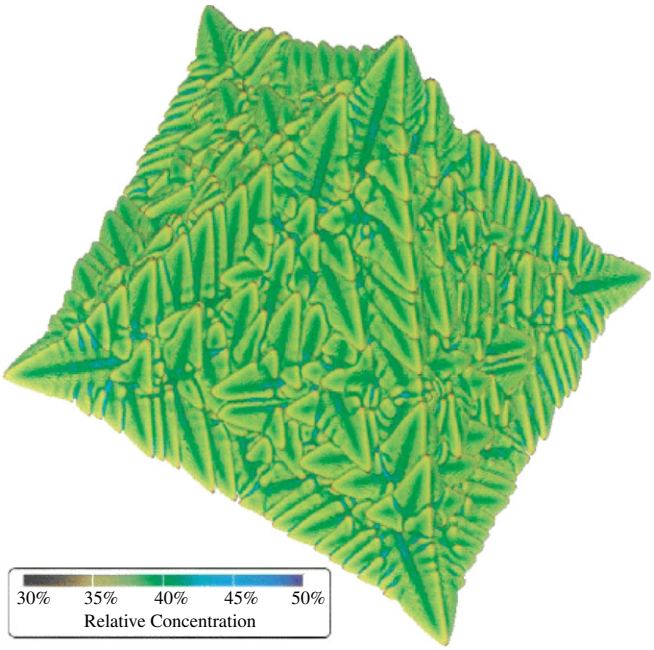


Figure 3. A 3D dendrite solidified out of highly supersaturated binary melt under isothermal conditions. Numerical values of the solute concentration at the solid–liquid interface are indicated in the map on the left hand side (courtesy of George and Warren [115]). Reprinted from [115], copyright (2002), with permission from Elsevier.

time-stepping schemes. It was demonstrated that at high cooling rates the supersaturation is replaced by the thermal undercooling as the driving force for growth. By using an adaptive finite volume method Lan *et al* [116] performed simulations of free dendritic growth in large computational domains, so that the temperature field was not affected by the domain boundaries.

Even though realistic microstructural patterns were obtained in [38, 39, 116], the models still exhibit interface thickness dependent results and the presence of solute trapping. Tiaden *et al* [117] proposed to solve this problem by introducing two separate concentration fields, one for the solid and one for the liquid, and by interpreting the interface as a mixture of two phases. Later Kim *et al* [118] demonstrated that the requirement of a local equilibrium between the two phases allows one to eliminate one of the concentration fields. The resulting model has a surface tension that is independent of the interface thickness and can be used for arbitrary phase diagrams; however, some non-physical effects remained, in particular, surface diffusion [118].

Karma [33] re-examined the existing phase field models for binary alloys in the thin interface limit. He demonstrated the presence of solute trapping in the low growth regime due to the jump in the chemical potential across the interface and proposed a new model for solidification of a dilute binary alloy under the assumption of vanishing solute diffusivity in the solid. The model includes a phenomenological 'antitrapping' solute current in the mass conservation relation. This antitrapping current counterbalances the solute trapping effect and provides the freedom to eliminate kinetic undercooling as was done for the solidification of pure melts [31, 32].

A rigorous derivation of the model and a thin interface analysis was presented by Echebarria *et al* [119]. The authors also gave physical interpretations of non-equilibrium effects arising at the phase boundary when mesoscopic interface thicknesses were used.

Ramirez *et al* [120] extended [33] to the non-isothermal case and demonstrated good agreement with one-dimensional steady-state analytical solutions. Thermosolutal dendritic growth was simulated in two dimensions for vanishing solutal diffusivity in the solid and for Lewis numbers (ratio of the liquid thermal to solutal diffusivities) up to 50. However, their results become inaccurate when the diffusivities differ significantly [121]. Ramirez and Beckermann [122] studied the effects of the alloy composition, Lewis number and undercooling on the selection criterion for the operating point of the dendrite tip. They found that the selection theory works well in the case of pure and solutal melt, but breaks down for low concentrations, where both thermal and solutal effects are important. Lan and Shih [123, 124] demonstrated the effect of the antitrapping term on dendritic growth. Namely, they found that the tip radius and the velocity for the dendritic morphology calculated with large interface thicknesses (50 nm) combined with the antitrapping term are in very good agreement with simulations performed with a small interface thickness (2 nm) and no antitrapping term present. However, the concentration profiles obtained in these two cases differ significantly.

Haxhimali *et al* [125] investigated atypical growth directions of binary alloys. It was demonstrated that primary dendrite growth directions can vary continuously between different crystallographic directions as a function of composition dependent anisotropy parameters.

A phase field model based on a Gibbs energy functional was proposed by Loginova *et al* [50]. The model reproduces the following important types of $\gamma \rightarrow \alpha$ phase transitions in the Fe–C system: from C diffusion controlled growth through the Widmanstätten morphology to massive growth without partitioning of carbon. By introducing highly anisotropic surface energy, two-dimensional side plates (the so-called Widmanstätten plates) emanating from an austenite grain boundary were simulated. Growth of such a colony of plates is shown in figure 4. It was found that the growth of the plates depends on the relation between the anisotropy of the surface energy and the interface thickness. When the interface thickness is close to its physical value of 1 nm, the strength of the anisotropy must be taken in the order of 100 to observe the growth of such precipitates. A model for a non-dilute Ti–Al binary alloy showing transitions from diffusive to massive growth for $\alpha \rightarrow \gamma$ precipitations was presented by Singer *et al* [126]. The model is coupled to the thermodynamic database ThermoCalc, using the thermodynamically correct values of the free energies in the α- and γ-phases.

In rapid solidification simulations of binary alloys by Fan *et al* [127] it was shown that the dendrite tip velocity exhibits a transition between two different regimes with increasing undercooling. Simulations with low undercoolings are in accordance with the kinetics of the Stefan problem where the interface is in a local equilibrium. However at high undercoolings the tip velocity follows a linear dependence on the undercooling.

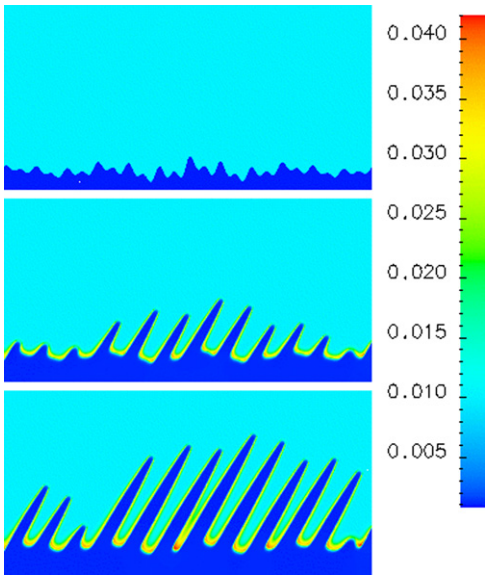


Figure 4. A colony of Widmanstätten plates precipitating from an austenite grain boundary in an Fe–C alloy [50] calculated by the authors. Numerical values of C concentration are indicated in the map on the right hand side. Growing ferrite has a low C concentration whereas the parent austenite has a high C concentration.

3.1.3. Multicomponent systems. Early phase field modeling concentrated on the solidification of pure and binary alloys. However, most of the technologically relevant alloys consist of more than two components. In comparison with binary alloys, it is considerably more difficult (often impossible) to construct analytical expressions for thermodynamic quantities, e.g. molar free energy. Besides, some of the phase field parameters, such as interfacial mobility, cannot be related in a simple way to physical quantities. The usual approach to solve this problem is to couple the phase field equations to thermodynamic databases and equilibrium calculations. The thermodynamic quantities (e.g. molar Gibbs energy, chemical potential, concentration) at the solid/liquid interface are calculated from thermodynamic databases and then imposed on the phase field equations. Even though promising steps have been made towards modeling of multicomponent phase transformations, most of the published results are restricted to ternary alloys (often under dilute approximation). Another commonly used simplification is that off-diagonal components of the diffusion matrix are neglected.

Suzuki *et al* [128] extended the phase field model for dilute binary alloy solidification [118] to ternary alloys. They presented isothermal free dendritic growth for Fe–C–P alloys and studied the effect of the ternary alloying element on the secondary arm spacing. Kobayashi *et al* [129] proposed a phase field model for ternary alloys, which incorporates the thermodynamic data via an *a priori* calculated tie-line map of the ternary phase diagram. They performed 2D simulations of dendritic growth under the condition of a vanishing kinetic coefficient. Chen *et al* [130] reported on the direct linking of the the Ti-based thermodynamic database from CompuTherm and the kinetic software DICTRA to a two-phase field model for modeling the growth of precipitates in the ternary Ti–Al–V system.

By employing the number of moles as a concentration variable, Cha *et al* [131] presented a phase field model for the solidification of multicomponent alloy systems with both substitutional and interstitial elements. They studied equilibrium and kinetic properties of isothermal solidification of ternary alloy systems in one dimension. However, their model had severe limitations on the interface thickness due to the variation of the chemical free energy in the interface region. Recently, Cha *et al* [132] modified this model to relieve the interface thickness limiting constraints by defining the interfacial region as a mixture of the solid and liquid phase with the same chemical potential and determining the phase field parameters at the thin interface limit. They presented 2D dendritic growth in the case of a non-dilute ternary alloy. A phase field simulation for solidification in ternary alloys using the finite element method has been reported by Danilov and Nestler [133].

In the model for multicomponent solidification presented by Qin and Wallach [134] the concentration of solutes in the forming solid phase can be fully determined from thermodynamic data. A detailed description on the model derivation, coupling to the thermodynamic database MTDATA and the calculation of the phase field mobility for the binary Al–Si system are given in [134]. The model was applied for the simulation of morphological transitions in semisolid metals [135]. Wu *et al* [136] simulated the evolution of interdiffusional microstructures in an Ni–Al–Cr alloy by calculating the free energy and mobility data with the CALPHAD approach. An overview on the combination of the CALPHAD techniques with phase field models is given by Steinbach *et al* [137].

There have been a few attempts to model multiphase transformations in multicomponent alloys. Grafe [138] extended the multiphase field model [139, 140] for unary alloys to general multicomponent alloy systems. The model was coupled to the thermodynamic database ThermoCalc to simulate the growth of ferritic dendrites and the subsequent formation of austenite in the ternary Fe–C–Mn alloy during directional solidification [34]. The multiphase approach has also been extended by Qin *et al* [141] to multicomponent systems and has been incorporated with a thermodynamic database. Simulations of Al–Si–Cu–Fe alloys with four possible forming solid phases predicted microstructures, which agree with experimental observations. Kim [142] extended the antitrapping current used in binary alloys to non-dilute multicomponent alloys and found that multicomponent effects can be expressed in a concise matrix form. Boettger *et al* [143] showed that phase field models can be applied to the simulation of equiaxed solidification in technical alloys such as the commercial Mg–Al–Zn alloy AZ31, a hypereutectic Al–Si–Cu–Mg–Ni piston alloy and AlCu4Si17Mg. The complex phase diagrams were calculated by a local quasi-binary extrapolation.

3.1.4. Effects of natural convection and growth under shear flow. Modeling of solidification is not complete without taking into account convection in the melt. Convection modifies the diffusion fields, the kinetics of the interface and the number of defects in the solid and can lead to the formation of completely different growth patterns in comparison with purely diffusive growth. Coupling the phase field and Navier–Stokes equations involves different time and length scales. Solving the coupled system is a non-trivial task, which requires sophisticated numerical techniques and often the use of adaptive grids.

Tönhardt and Amberg [144, 145] proposed to treat both liquid and solid phases as Newtonian fluids with the viscosity of the solid much higher than that of the liquid phase. They studied the effects of shear flow on the evolution of a small nucleus attached to a wall in two dimensions. It was demonstrated that the flow, directed parallel to the wall, significantly alters the shape of a dendrite from the original 4-fold symmetry. In particular, the main stem growing perpendicularly to the flow tilts increasingly towards the upstream direction, where the thermal field is suppressed and the fresh melt has a lower undercooling. Also, the side branches on the upstream side of the dendrite are promoted, while on the downstream side they are inhibited. Tönhardt and Amberg [146] also studied the effects of natural convection on succinonitrile crystals in extremely large domains. When a single dendrite grows, it releases latent heat and heats the surrounding liquid. Due to convection in the melt, the flow is directed from below upwards and at some point is forced to turn downwards again. This gives rise to two vortices far above the dendrite. The size of the vortices is an order of magnitude larger than the dendrite's size and depends on the viscosity of the liquid and the domain size [146]. However, in the vicinity of the dendrite the flow can be approximated as uniform in the upstream direction. Since resolving large computational domains is extremely costly, the phase field studies reviewed below utilize the approximation of the imposed forced flow with downward direction.

Beckermann *et al* [30] incorporated melt convection phenomenologically in a standard phase field model for pure and binary alloy solidification. Whereas the phase field equation was left unchanged from the purely diffusive case—assuming that the phase field is not advected by the flow—the Navier–Stokes equation is modified with a distributed dissipative interfacial drag term to model the non-slip condition at the solid–liquid interface. The model was applied to study dendritic growth at high undercoolings with emphasis on the operating point of the tip [30, 147] and Ostwald ripening of a binary alloy solid/liquid mush in 2D [148]. It was found that the presence of convection increases the growth rate of the mean radius of the solid particles. Anderson *et al* [149] derived a phase field model of a pure substance with convection, which can be reduced to a non-equilibrium form of the Clausius–Clapeyron equations and studied the sharp interface limits of the model [150]. They investigated the solidification of a planar interface under the density change flow and shear flow. Tong *et al* [151] incorporated thermal noise into the model of Beckermann [30] to study the dendritic shape and sidebranching in the presence of a forced flow. It was demonstrated that due to the strong effect of the direction of the flow relative to the growth axis the growing dendritic crystal assumes a highly asymmetric shape. The flow promotes

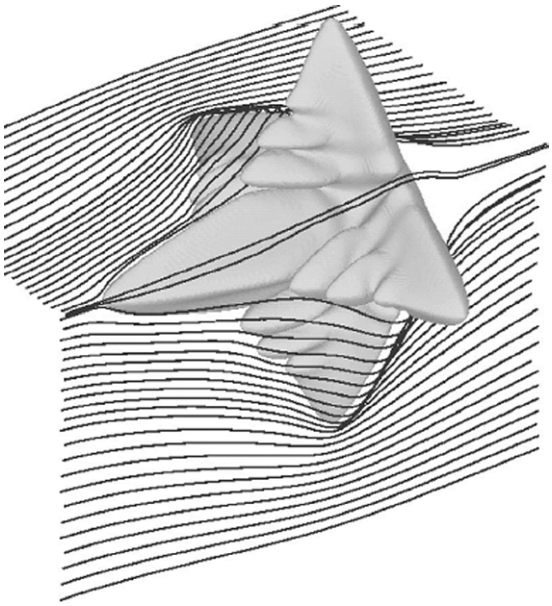


Figure 5. A freely growing dendrite in a forced flow. The lines represent computed stream traces directed from left to right. Reprinted with permission from [80]. Copyright 2001 by the American Physical Society.

the growth of the arm directed upstream to the flow, has a small effect on the arms growing normal to the flow and damps the development of the arm growing in the downstream direction. Convection is also found to increase the amplitude and frequency of side branches along the upstream growing dendrite arm.

Using adaptive finite-element grids Jeong *et al* [80] simulated growth of a 3D dendrite in a forced flow. The melt convection has a strong influence on the 3D dendrite growth morphology, as presented in figure 5. The mechanism by which the flow alters the growth of a dendrite is inherently different in two and three dimensions. When the dendrite grows perpendicular to the flow direction, the transfer of heat and solute in 2D must flow upwards and over the dendrite. In contrast, in 3D, the fluid may flow horizontally around a vertical main branch, thus providing an efficient transfer from the upstream to the downstream side of the dendrite. Similar 3D simulations were independently performed by Lu *et al* [152].

At low undercoolings, the dendrite tip reaches its steady state very slowly. As demonstrated by Lan *et al* [153, 154] in the case of pure diffusive growth at undercooling 0.1, the dendrite does not approach the steady state even at very large computational times. This could be due to the disturbance of the thermal field by thermal diffusion of growing side branches [155]. In contrast, the presence of forced flow increases the tip velocity and the dendrite arm growing in the upstream direction approaches a constant velocity much more rapidly. Besides, the thermal effects of the side branches are suppressed by the flow. By performing 2D simulations for various undercoolings, Lan *et al* [154] demonstrated that the rescaled dendrite tip shape is unaffected by the flow and the undercoolings and remains parabolic. Their calculated results agree well with the Oseen–Ivantsov solution, which is a modification of the Ivantsov solution in the presence of flow.

Non-isothermal free dendritic growth [123, 124] and directional solidification [156] of a binary alloy in a forced flow were investigated by Lan *et al* and effects of the anti-trapping current and the interface thickness were reported. Conti [157, 158] studied the influence of the advection flow due to different densities of the solid and liquid phases. He found that at fixed undercoolings, as the ratio of solid to liquid density increases, the tip velocity decreases and the tip radius increases. Miller *et al* [159, 160] derived a discrete phase field model coupled with the lattice Boltzmann method for computing the melt convection. The method was applied for studying the influence of shear flow on dendritic and seaweed growth [161, 162]. Qin and Wallach [135] reported on the effects of lamellar and turbulent flow on the growth morphologies in semisolid metal processing. While dendritic growth is encouraged under conditions of a pure laminar flow, the growth morphology changes from dendritic to rosette and finally to spherical in the presence of turbulence.

Convective effects have been taken into account for rapid solidification in pure supercooled metallic melts of levitated droplets by Galenko *et al* [163, 164] and Herlach and Galenko [165] (and references therein). It was found that theoretical results and sharp interface calculations deviate systematically from the experimental results. By means of a phase field model incorporating forced convective flow and a small amount of impurities it was shown that the solidification front's velocity is increased.

3.1.5. Noise, fluctuations and the potential for nucleation in the phase field model. Dendrites have complex shapes due to the appearance of secondary side branches behind the growing tips of the primary stalks [166, 167]. The physical origin of the sidebranching is small noise perturbations, initially localized at the tip. Amplification of these perturbations to a macroscale along the sides of a steady-state needle crystal leads to the development of side branches. The side branches consequently appear behind the tip, which implies the presence of a continuous source of noise at the tip. Experimental results [168] indicate that thermal noise, originating from microscopic thermal fluctuations inherent in the bulk matter, is responsible for the appearance of side branches.

Typically noise is included into phase field models in a rather ad hoc manner. Kobayashi [16] introduced noise by adding a term which is evaluated using a random number generator and showed that sidebranching strongly depends on the strength of this noise. Other studies [36, 38] used a similar technique by including a noise term into the phase field equation, thus simulating fluctuations only at the interface. However, these interfacial fluctuations give rise to side branches only in a system with high driving forces (high undercooling and/or supersaturation).

Pavlik and Sekerka [169, 170] derived stochastic forces due to thermodynamic fluctuations for anisotropic phase field models. Based on general principles of irreversible thermodynamics they showed that the stochastic forces are anisotropic. González-Cinca *et al* [171] studied the

appearance of side branches of solutal dendrites by means of the phase field method. Karma and Rappel [172] incorporated thermal noise quantitatively into a phase field model for pure solidification. They related the amplitude of noise to physical quantities. Two types of noise were considered in [172]: the non-conserved interface noise originating from the exchange of atoms between the two phases and conserved bulk noise originating from fluctuations in the heat current in the solid and liquid phases. They demonstrated that for typical growth conditions at low undercooling the conserved noise is the most relevant one. The long-wavelength interface fluctuations driven by the conserved noise amplify to a macroscopic scale by the morphological instability on the sides of the dendrite. In contrast, the non-conserved noise in the evolution equation for the phase field variable drives short-wavelength fluctuations that are damped and do not affect sidebranching. Consequently, at low undercoolings the non-conserved noise can be left out to speed up computations.

Theoretically, purely deterministic phase field simulations of dendritic growth should produce needle-like dendrites without side branches. However, if the diffuse interface region is not properly resolved, e.g. a coarse finite difference mesh is used, calculations will usually exhibit sidebranching typical for real dendrites as discretization errors introduce noise into the calculations.

Solidification of an undercooled liquid begins with a fluctuation which produces a cluster of atoms. If the size of the cluster exceeds a critical size, the cluster becomes a stable nucleus, which later develops into a crystal. The nucleation process can be incorporated into the phase field method through modeling thermal fluctuations with Langevin noise. The amplitude of the noise is determined by the fluctuation–dissipation theorem. This approach has been used for modeling nucleation in pure melts [173] and binary alloys during eutectic solidification [174]. However, modeling Langevin noise is extremely time consuming and thus not practical. Instead, in order to initiate crystallization, nuclei are placed randomly into the computational domain [175]. Gránásy *et al* [176, 177] extensively studied the nucleation process in binary alloys. They calculated the height of the nucleation barrier and investigated competing nucleation, when a large number of crystalline particles form simultaneously and compete with each other.

3.2. Directional solidification

In directional solidification experiments, a sample containing an alloy is pulled at a constant velocity in an externally imposed temperature gradient. The solidification front follows the direction of the applied thermal gradient. The resulting microstructures depend on the type of the alloy, the ratio of the pulling velocity and the temperature gradient. In the case of dilute binary alloys, cellular or dendritic patterns are formed during the single solid phase transformations, as demonstrated in figure 6. For non-dilute alloy concentrations close to the eutectic point, two stable solid phases of different compositions can grow from a metastable liquid. These processes are called eutectic or peritectic phase transformations. For off-eutectic compositions, the coexistence between dendrites and

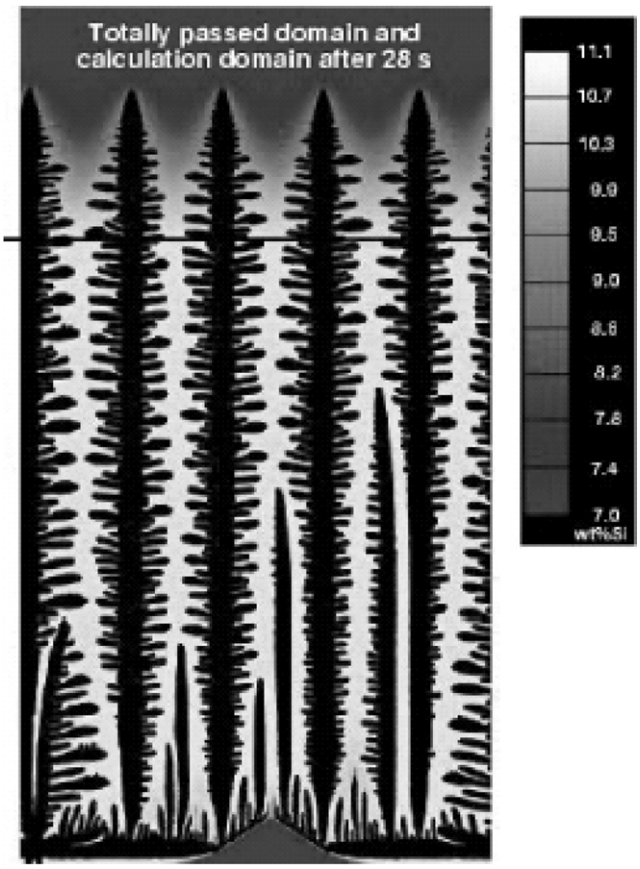


Figure 6. Simulation of dendritic growth and selection of primary spacing during directional solidification of an Al–Si alloy. The solute concentration in solid is darkened. Side branches are provoked by numerical instabilities (courtesy of Diepers *et al* [180]). Reprinted from [180], copyright (2002), with permission from Elsevier.

coupled growth is also observed. While eutectic and peritectic solidification is discussed in section 3.4, here we present directional solidification of a single solid phase. According to Mullins–Sekerka theory of morphological instabilities [52, 53], the interface morphology varies during directional solidification of a binary alloy from planar growth at low velocities (lower than the critical velocity) to cells and then to dendrites, which become finer and finer until they give rise to cellular structures again when the velocity is close to the limit of the absolute stability. If the velocity is higher than this limit a planar interface is stabilized. Numerous phase field studies on directional solidification are devoted to understanding these fascinating morphological transitions. Typically, effects of thermal gradients and pulling speeds on the selection of the growth morphology and its wavelength are investigated. Most of the reported simulations on directional solidification are limited to two dimensions. To our knowledge only Plapp [178] has studied 3D directional solidification so far.

Boettinger and Warren [111] used the binary phase field method [38] with a frozen temperature approximation to simulate the transition from cells to planar growth at high solidification rates. Kim and Kim [179] presented simulations of banded structures during rapid directional solidification under a constant thermal gradient. These structures are

alternating patterns of the partitioned cellular/dendritic and partitionless planar front structures. Diepers *et al* [180] studied qualitatively primary dendritic spacing during directional solidification of the Al–Si alloy. They varied the cooling rates in time and found that the spacing assignment is affected by the history of the microstructural development. Seol *et al* [181] investigated hardening of the mushy zone during directional solidification of carbon steels. Bi and Sekerka [182] concentrated their interest in the transition from a planar interface to shallow cells, which occurs at low pulling velocities near the onset of the morphological instability. The observed wavelength selection occurs by means of tip splitting or coarsening. It was demonstrated that the cell shapes become nearly sinusoidal when the solid/liquid interface approaches steady state. Provatas *et al* [183] introduced anisotropy of the surface energy to a phase field model for directional solidification. They demonstrated that at small anisotropies directed at 45° relative to the pulling direction, both seaweed and dendritic morphologies can appear. In order to classify the crystal morphology, a method based on the distribution of the local interface velocity was presented. The transition between seaweeds and dendrites was characterized as a function of thermal gradients and pulling velocities. Lan and Chang [184, 185] further applied the model presented in [111] to study the morphological instability [185] and cellular structures in the regime of low velocities and high temperature gradients, where both solutal boundary layer and cell wavelength are large [184]. They relieved the frozen temperature approximation by solving the heat equation [39] also and found that the effect of the latent heat release is not significant at high cooling rates. Echebarria *et al* [119] applied Karma's model [33] for low-speed directional solidification of a dilute binary alloy. They validated the model by calculating the Mullins–Sekerka stability spectrum and nonlinear cellular shapes for realistic alloy parameters. Meca and Plapp [186] studied the oscillatory instabilities and bifurcation behavior in cells closely above the Mullins–Sekerka instability threshold in dilute binary alloys in 3D. Different morphologies were found: hexagons, stripes and inverted hexagons. Greenwood *et al* [187] presented a scaling function for the wavelength selection of cellular patterns. The cell spacing was examined for a wide range of thermal gradients and alloy compositions and it was found that the scaling function is in remarkable agreement with experimental data. It was also revealed that the spacing of the primary branches reaches its maximum at intermediate values of the pulling velocities. Rasin and Miller [188] performed a numerical study of cellular morphology evolution during Czochralski growth in the Ge–Si system. The growth is characterized by low constitutional undercoolings and a large size of the resulting structures. It was found that the wavelength of the cellular structure depends weakly on the pulling velocity and strongly on the thermal gradient. Lan *et al* [189] reported on a quantitative study of deep cell formation in directional solidification in thin films. They proposed a simple interface model to correct the solute trapping introduced due to the use of the thick diffuse interface. The model restores the equilibrium segregation at the interface.

Pons *et al* [190] presented a method how to stabilize a given lamellar spacing by a heat pulse feedback control mechanism in front of the tip of the growing cells for temperature gradients, where naturally completely different lamellar spacings would occur. The results of the simulations were confirmed experimentally.

3.3. Grain boundary

In a cast there are millions of dendrites forming independently out of solution and subsequently interacting with each other through heat and solute diffusion. When the dendrites approach each other, they impinge and form grain boundaries. Within each grain the atoms are all aligned; adjacent grains however differ in orientation. The subsequent process of some grains shrinking and disappearing while others grow at their expense is called grain growth. Grain growth occurs because of the tendency to reduce the total grain boundary area and thus to reduce the associated surface energy. Hence, the mean grain size increases with time. The properties of a material are dependent on the size of its grains and therefore it is of some interest to gain a better understanding of this process. For grain growth the phase field methods have been developed along two independent routes: the first one is based on the work of Chen and Yang [191]; the second one is proposed by Kobayashi *et al* [192]. Below we discuss ideas and applications of the two approaches.

Chen and Yang [191] proposed a multiorder model describing grain growth, in which the grains of different orientations are represented by a set of non-conserved order parameter fields. In the model, N predefined orientations are allowed, and each of them is assigned to an order parameter. Consequently, N phase field equations must be solved for simulating grain growth. When the number of allowed grain orientations is smaller than the number of grains, there is a finite probability for grain growth to occur not by boundary migration but through coalescence of grains of the same orientation. As was verified by Fan and Chen [193] in 2D simulations, at least 100 distinct grain orientations are needed to avoid a significant amount of coalescence. Later Krill and Chen [194] proposed a dynamic grain-orientation reassignment procedure, which allows one to reduce the number of grain orientations to 25 and performed impressive 3D simulations of grain coarsening reprinted in figure 7. The model proposed by Chen has been modified and used extensively in the area of solid–solid phase transformation. Kazaryan *et al* [195–197] and Suwa *et al* [198] studied the influence of grain boundary energy anisotropy and mobility on grain growth; Fan *et al* investigated the effects of solute drag on grain growth kinetics [199] and Ostwald ripening in two-phase systems [200, 201]; Ramanarayan and Abinandanan [202–204] reported on the effects of the grain boundary evolution on spinodal decomposition in polycrystalline alloys. Other examples include texture evolution [205], sintering [206] and interaction of grain boundaries with small incoherent second-phase particles [207]. The solute drag on moving grain boundaries in a binary alloy system was studied by Cha *et al* [208].

A model, similar to the one of Chen, the so-called multiphase field model, for grain growth was proposed by

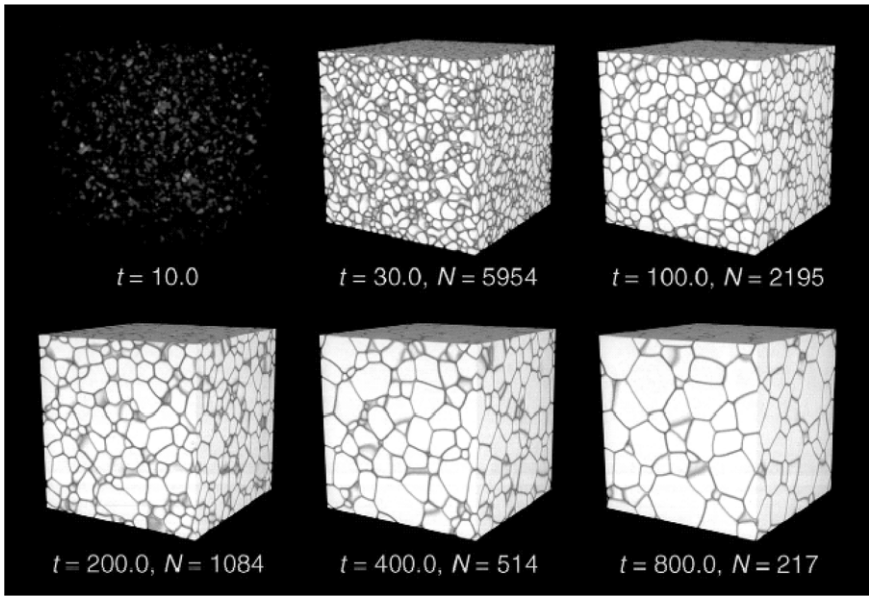


Figure 7. Grain coarsening process obtained from 3D simulations of grain growth. In the model the grain boundary energy and mobility is isotropic. The elapsed time t and the number of grains N are specified for every image. The microstructure at $t = 10.0$ illustrates the homogeneous nucleation of crystallites from the undercooled liquid initial state (courtesy of Krill and Chen [194]). Reprinted from [194], copyright (2002), with permission from Elsevier.

Steinbach *et al* [139]. In this model, a special constraint is imposed, requiring that the sum of all order parameters in every point must be 1, which implies that the order parameters represent the volume fraction of grains of different orientations. The model has been applied to study grain growth in thin metallic films [209, 210], transient growth of dendrites [90] and has been used extensively for eutectic and peritectic transformations (see section 3.4).

In the phase field model proposed by Kobayashi *et al* [192, 211–215] the microstructure is characterized by two order parameters: one describing the crystalline order and the other reflecting the crystalline orientation of each individual grain. Whereas the relaxation of the crystalline orientation parameter simulates grain rotation, which is absent in the multiorder parameter models, this order parameter is undefined in a disordered liquid phase. Different from multiorder parameter models for grain growth, the model is invariant under rotation of the reference frame and can accommodate any number of grain orientations. The model was extended to simultaneously simulate solidification of arbitrarily oriented dendrites, their impingement and subsequent coarsening in pure [216] and binary melts [217]. Lobkovsky and Warren [218–220] analyzed sharp interface limits of [192, 211] and studied the dynamics of grain boundaries, rotation of the grains and premelting of high-angle boundaries. Bishop and Carter [221] proposed to couple the results of atomistic simulations to the grain boundary phase field model. Despite the conceptual simplicity of the model [192, 211] in two dimensions, the extension to three dimensions is not straightforward and is an active area of research [222, 223].

A computationally efficient scheme for calculating ideal grain growth of thousands of grains was introduced by Kim *et al* [224]. It was demonstrated that the presented model reproduces the Neumann–Mullins law for individual grains under coalescence-free conditions. The steady-state grain size distribution in 3D appeared to be very close to the Hillert 3D distribution independent of the initial conditions of grain sizes.

Grohagen and Ågren [225] studied the dynamics of grain boundary segregation to a stationary boundary as well as the solute drag on a moving boundary by introducing a concentration dependent height of the double-well potential of the free energy.

By allowing fluctuations of the crystalline orientation in the liquid phase, Gránásy *et al* [176, 177] incorporated nucleation of solid particles into the phase field model [192]. They also presented a phase field model for polycrystalline growth in which new grains nucleate at the solidification front [226]. The presence of particulates in the liquid or a small rotational–translational mobility ratio gives rise to a rich variety of polycrystalline growth patterns, from dendrites through seaweeds to needle crystals. For a comprehensive review on modeling polycrystalline solidification the reader is referred to Gránásy *et al* [227, 228]. A theory of heterogeneous nucleation was presented by Gránásy *et al* in [229].

3.4. Three-phase transformations: eutectic, peritectic and monotectic

So far we have considered the phase transformation involving only two phases: the parent liquid phase transformed into the solid phase. However, in practice, most phase transformations involve multiple phases. Though the multiphase field models reviewed in this section are rather general, in that they are valid for most kinds of transitions between multiple phases, their practical application is narrowed to the three-phase eutectic and peritectic transformations, which are observed during directional solidification of binary alloys.

During a eutectic transformation, two stable solid phases of different compositions may grow from an undercooled

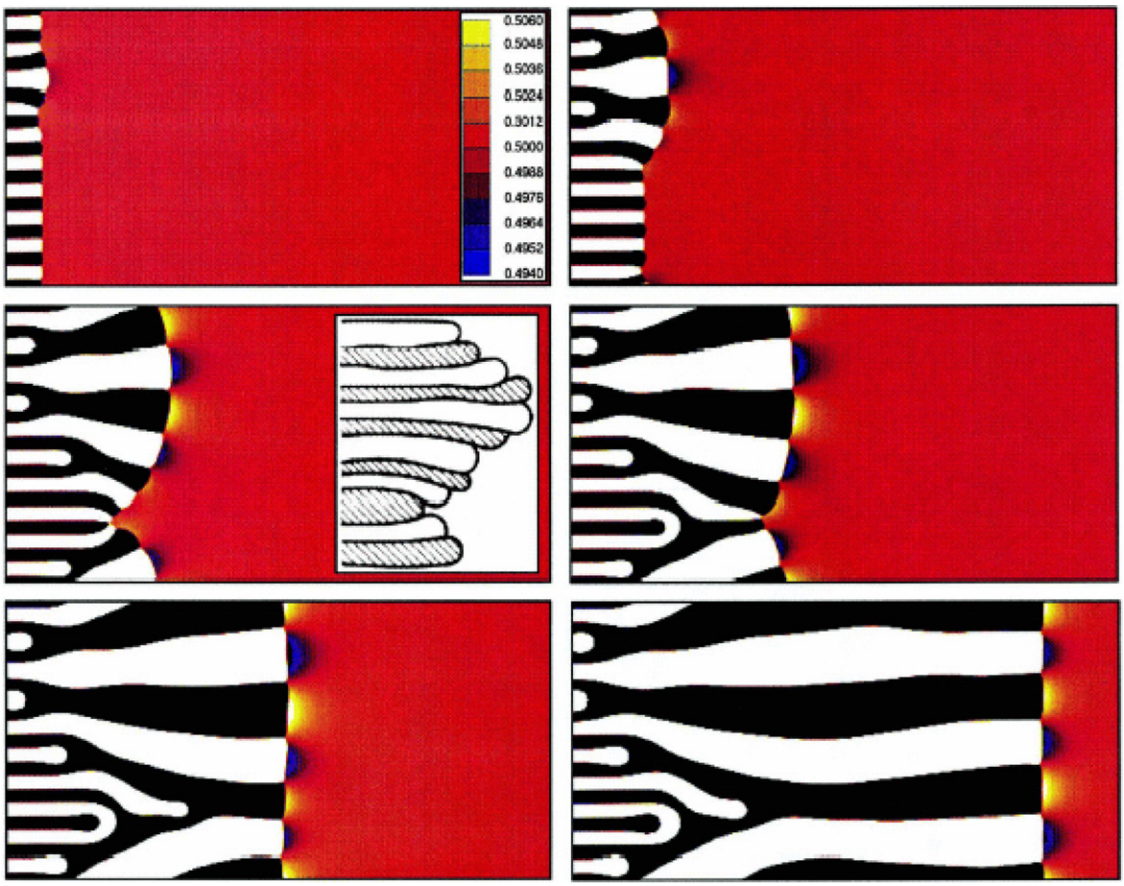


Figure 8. Evolution of 2D eutectic lamellae of a binary alloy and the selection process of lamellar spacing. The black and white colors represent the two solid phases, and the numerical values of the solute concentration in the liquid are indicated in the map in the first picture (courtesy of Nestler and Wheeler [233]). Reprinted from [233], copyright (2000), with permission from Elsevier.

liquid phase when its temperature is below the eutectic temperature. The two solid phases advance in a steady-state manner and form a periodic structure of lamellae or rods with one of the solid phases embedded in a matrix of the other solid phase. Depending on the relation between capillarity and diffusive bulk transport between adjacent solid phases more complex patterns such as bifurcations, limit cycles and spatio-temporal chaos may also arise. In peritectic alloys, the primary solid phase growing from the metastable liquid usually possesses a dendritic morphology. Below the peritectic temperature the new (peritectic) solid phase nucleates at the solid–liquid interface. The peritectic solid phase grows around the parent phase until either the primary phase is completely melted or is entirely engulfed within the new phase.

A first phase field model for multiphase transitions was proposed by Steinbach *et al* [139]. In the model, each of the N phases is identified by a phase field variable $\phi_\alpha(\boldsymbol{x}, t)$, $\alpha = 1 \ldots N$ and at every point in the system the sum of all ϕ_α must equal 1. The free energy functional is constructed to account for pairwise interactions between the different phases. Nestler and Wheeler [230] included surface energy and interface kinetics anisotropy into the multiphase field method [139]. Later, Steinbach and Pezzolla [140] reformulated [139] in terms of interface fields to allow for the decomposition of the nonlinear multiphase field interactions into pairwise interactions of the interface fields. The sharp interface limits of these multiphase field models were studied in [230, 231], where the classical laws at interfaces and trijunctions were recovered. Tiaden *et al* [117] extended the model of Steinbach [139] to incorporate solute diffusion into the multiphase field approach. The model was applied to simulate peritectic solidification in the Fe–C alloy for isotropic ferritic particles and during directional solidification, when the ferrite phase grows in the dendritic morphology [232]. In this case, the austenite grows around the undercooled part of the ferritic dendrite, consuming both the liquid and the ferrite.

In contrast to multiphase field concepts, the phase field model of eutectic growth proposed by Drolet [174] contains only one phase field variable for distinguishing solid and liquid phases. A conserved concentration field defined relative to the eutectic composition distinguishes the two solid phases. The model was used to study directional solidification and lamellar eutectic crystallization under isothermal conditions.

Further developing the multiphase field method, Nestler and Wheeler [233] reported computations of a wide range of realistic phenomena arising during eutectic and peritectic phase transformations. They studied the selection process of eutectic lamellar spacings depending on the initial spacing. On too small initial spacings, the competitive growth of neighboring lamellae leads to the annihilation of some lamellae with a subsequent increase in the spacing of the remaining lamellae. This process is demonstrated in figure 8. In

the case of a peritectic alloy, they computed the dissolution of the primary solid and the simultaneous growth of the peritectic phase. Numerical results of the growth of a peritectic solid over a planar solid–liquid interface were presented and it was demonstrated how the new phase surrounds and engulfs circular particles of the primary phase. The multiphase field concept was also applied to study the monotectic transformation. This phase transformation is commonly observed in a wide class of alloys with a miscibility gap. In monotectic reactions the parent liquid phase decomposes simultaneously into a solid phase and another liquid. By incorporating Navier–Stokes equations to take into account melt convection, Nestler and Wheeler [234] presented simulations of lamellar monotectic growth structures, which exhibit wetting phenomena, coarsening and particle pushing. Later the model was extended for a general class of binary three-phase alloy systems and was also applied to describe grain growth phenomena [210].

Lo *et al* [235] considered directional solidification of peritectic alloys under purely diffusive growth conditions. In this regime (large thermal gradients and low pulling velocities) both solid phases are morphologically stable and the interface dynamics is controlled by a subtle interplay between the growth and nucleation of two competing solid phases. Morphological transitions from islands to bands and the dynamics of the peritectic phase spreading on the parent phase after nucleation were simulated in two dimensions. These simulations show good qualitative agreement with experimental observations [236]. Apel *et al* [237] presented a multiphase field model for eutectic growth and performed simulations in two and three dimensions. In two dimensions, lamellae exhibit either stable or unstable oscillating growth depending on the initial lamellar spacing. In three dimensions, the fibrous growth structure appears when the imposed phase fraction ratio is large. The fibers tend to form a hexagonal arrangement with increased growth velocities.

Plapp and Karma [238] presented a phase field study of binary eutectic solidification in the presence of a dilute ternary impurity. They demonstrated growth of eutectic two-phase cells, also known as eutectic colonies. It was found that lamellae do not grow perpendicularly to the envelope of the solidification front. This has an important quantitative effect on the stability properties of the eutectic front. The large-scale envelope of the two-phase cells does not converge to steady state, but exhibits cell elimination and tip splitting, which is in agreement with experimental observations. Akamatsu *et al* [239] studied the stability of lamellae and the trijunction motion during eutectic solidification by experiments and phase field simulations. They found that the lamellar eutectic growth is stable for a much wider range of spacings than that predicted by the stability analysis. The discrepancy is explained by the growth of lamellae which are not exactly normal to the large-scale envelope of the composite interface, as it is assumed in the theories.

Kim *et al* [240] presented a eutectic phase field model with isotropic and anisotropic interface energies and kinetic coefficients. The isotropic model was used to simulate two-dimensional lamellar eutectic growth in binary eutectic alloys at eutectic and hyper-eutectic alloy compositions. They demonstrated that at the hyper-eutectic composition the basic lamellar structures transform to oscillatory and tilted patterns and that the characteristic scales of the patterns agree quantitatively with experimental results. Toth and Gránásy [241, 242] studied the nucleation behavior in eutectic systems of fcc structures by applying the multiphase field approach [139] and the phase field model for polycrystalline growth [176, 177, 226]. Lewis *et al* [243] performed a study of a eutectic binary system. They investigated transitions between rods and lamellar structures as a function of the volume fraction of the minor phase, as well as spacing selection via rod branching. They performed simulations of equiaxed eutectic solidification in 2D and lamellar eutectic structure in 3D (see figure 9). Folch and Plapp [244, 245] presented a three-phase field model for quantitative simulations of low-speed eutectic and peritectic solidification. The design of the model allows one to separate the dynamics of the solid–liquid interfaces and the trijunction points and to eliminate all unphysical thin interface effects for each of the solid–liquid interfaces. This non-variational model also includes an antitrapping current previously developed for binary single-phase solidification [33]. The model was applied to study the motion of the trijunction points and the morphological stability of 3D structures [246]. Green *et al* [247] studied the breakdown from eutectic growth into a single-phase dendritic front.

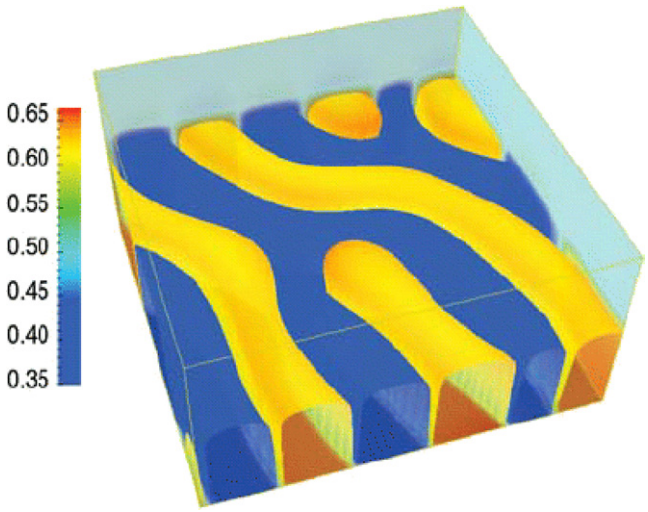


Figure 9. 3D eutectic growth. The snapshot demonstrates the area of the solid–liquid interface, which is obtained as an isosurface of the phase field variables. Numerical values of the solute concentration mapped to the surface are indicated in the map on the left hand side (courtesy of Lewis *et al* [243] (figure 1), ©The Minerals, Metals and Materials Society (TMS), Warrendale, PA, USA, 2004).

3.5. The phase field crystal method (PFC)

Elder *et al* [248, 249] proposed a new phase field theory, which allows for the atomistic modeling of the solid–liquid transition on diffusive time scales. The model naturally incorporates crystal anisotropy and elastic and plastic deformations. Among the first applications are the atomistic

study of grain boundaries, cracks, epitaxy [249] as well as eutectic solidification and dendritic growth [250].

The general idea of the phase field crystal model (PFC) is to consider time averaged positions of atoms leading to a density description of the (pure) material. This PDE description on an atomistic view allows the observation of morphological structure formation processes on diffusive time scales, which are not accessible by molecular dynamics simulations. The density is periodic in the crystalline phase and uniform in the liquid phase. A free energy functional is postulated, which is minimized by a periodic solution. The terms in the free energy functional describe the dependence on the undercooling, an operator fit to an inverse structure factor and a non-linear term, which enforces the lattice structure (fcc/bcc/hcp). The equations of motion are derived with the Cahn–Hilliard formalism in the same way as the traditional phase field equations. From the point of view of pattern formation theory the resulting equation is the conserved version of the simplest form of the Swift–Hohenberg equation (using only the cubic non-linear term).

The most interesting aspects of the model are the fact that multiple lattice orientations emerge naturally and that elasticity and dislocations are automatically included. This is an advantage over traditional phase field models, where these properties are *a priori* coarse grained and have to be re-inserted at considerable cost (grains, e.g., Kobayashi *et al* [192], elasticity, e.g., Uehara *et al* [251]). The derived equations of the PFC model contain sixth-order spatial derivatives leading to a 49-point stencil when discretized with finite differences. While a direct simulation is possible and was used in [249], the allowed time steps are rather small. Goldenfeld *et al* [252–254] and Athreya *et al* [255] proposed therefore an approach to solve the equations on adaptive grids. This computationally efficient approach decomposes the order parameter, which describes the density profile, in a slowly varying amplitude and phase. The equations are rotationally covariant and are derived from the renormalization group. Singer *et al* [256] proposed a fast Fourier solver, based on an Eyre preconditioner, which allows for the efficient solving of the equations in the regime, where the whole domain has solidified already and coarsening is the only dynamics present. In this case speed-ups of two orders of magnitude were obtained. Proofs of the numerical method are given in Cheng and Warren [257].

The visualization of a large number of simulated atoms is rather tedious. While an iterative averaging approach as in [249] leads to visual coarse graining, all of the relevant atomistic information is lost. Additionally it is not straightforward to extract directly meaningful information from the simulated atomistic data. Singer and Singer [258] therefore introduced a 2D wavelet transformation, which allows the immediate extraction of grain boundaries, grain orientations, lattice defects and grain boundary angles as well as the grain size distribution.

Dislocations have been studied by Berry *et al* [259]. It was found that the natural features of dislocation processes such as glide, climb and annihilation emerge from the PFC model without explicit consideration of elasticity theory or construction of microscopic Peierls potentials. Elastic/plastic interactions were studied by Stefanovic *et al* [260, 261] by introducing a second order time derivative in the PFC equations, adding 'instantaneous' elastic interactions as traveling waves in the domain. More rigorously Manjaniemi and Grant [262] derived the nonlinear transport equations for isothermal crystalline solids using a density functional theory based free energy and the Poisson bracket formalism to study the phonon dispersion in the crystal. At the linear level it was shown that Fick's law breaks down without dislocation which is in agreement with Cahn's theory of stress effects on diffusion in solids. Achim *et al* [263] studied the phase diagram and the commensurate–incommensurate transitions of the PFC model of a two-dimensional crystal lattice in the presence of an external pinning potential. Their results are consistent with the results of simulations for atomistic models of adsorbed overlayers. Elder *et al* [264] presented a combination of PFC theory with the classical density functional theory of freezing. It was shown that there is a direct connection between the correlation functions entering density functional theory and the free energy functionals of the PFC model. From these connections a PFC model for binary solidification was derived.

While the PFC model provides the framework to simulate dislocations and vacancies, they represent only the probability density to find a vacancy at a given position. The evolution of an initial vacancy diffuses away as shown in [249]. In order to control the positions and movements of vacancies Singer and Singer [265] presented a coupling to kinetic Monte Carlo simulations for the propagation of vacancies in the PFC generated lattice, which lead to the formation of vacancy bubbles in the crystal.

3.6. Other applications

Over the past few years phase field models have been applied to a rich variety of physical phenomena. Additionally, conceptually new ideas of phase field techniques have been introduced. In the semisharp model of Amberg [266], the sharp change in energy takes place at a phase field contour, which is identified with the phase boundary. The accuracy of the method is, in principle, only limited by the grid resolution; however, the numerical implementation of the method is complicated. The analytical work of Greenberg [267] introduces a semi-diffuse interface. In contrast to the conventional phase field models, the phase field variable reaches the constant bulk values within the phase boundary.

Morphological pattern formation in thin films and on surfaces has received great attention in the phase field community, e.g. [268–270]. The phase field method was successful in predicting surface instabilities induced by stress [271–273] and strain [274]. Various aspects of epitaxial growth were discussed in [275–277], including nanoscale pattern formations of mono- and ternary epilayers [278–280] and island growth [281]. For hetero-epitaxial thin films studies of surface instability [282], dislocations [283] and quantum dot formation [284] were reported. Among other phenomena in the thin films investigated with the phase field method are spinodal decomposition [285, 286], structural evolution in ferroelectric thin films [287, 288] and crystallization of silicon

thin films [289]. Sintering and vacancy diffusion have been addressed in [290].

The phase field approach was applied to model two-phase flows [291, 292] and Hele–Shaw flows in different viscosity contrast regimes [293–297]. Effects of the elastic bending energy on the static deformation of a vesicle membrane were investigated in [298, 299]. Morphologies arising during solidification in a narrow channel were studied in [97, 300, 301]. An interesting phase field model of stress evolution during solidification of a thin rod was presented in [302]. Other examples of the phase field applications include modeling of crack propagation [303–308], electrochemistry [309–311], shape memory polycrystals [312], electrically and magnetically active materials [313] and wave propagation in anatomical models of the heart [314].

3.7. Scaling for quantitative modeling

The validation of a phase field model is performed by demonstrating that the model reduces to a free-boundary problem in the limit of vanishing interface thicknesses (W much smaller than any physically relevant length scale). From the computational point of view, the choice of the smallest physical length scale l_c is crucial as it sets up the limit for the grid resolution $\Delta x < W \ll l_c$. In the classical sharp interface analysis of the phase field models (e.g. [10, 103, 104]) the capillary length d_0 is chosen as the relevant length scale and the ratio $\epsilon = W/d_0 \ll 1$ is used as an expansion parameter. However, the use of a microscopic interface thickness is not suitable for quantitative simulations at low growth velocities.

Karma and Rappel [31, 32] suggested that the diffusion length D/V is a more relevant length scale in the phase field models of solidification and proposed to use an 'interface Peclet number' $p = WV/D$ as the expansion parameter. They presented a phase field model for solidification of pure melt with equal solid and liquid diffusivities, the so-called symmetric model. It was demonstrated that the model converges to a free-boundary problem in the thin interface limit $p \ll 1$ (i.e. the interface thickness W is much smaller than the diffusion length but not necessarily the capillary length). With a specific choice of the interpolation functions in the free energy functional the thin-interface asymptotic analysis yields an expression for the interface kinetic coefficient that contains a finite-interface thickness correction

$$\beta = a_1 \left[\frac{\tau}{\lambda W} - a_2 \frac{W}{D} \right], \tag{45}$$

where a_1 and a_2 are positive constants of order unity, τ is the interface attachment coefficient, D is the diffusion coefficient and the dimensionless parameter $\lambda = a_1 W/d_0$ controls the strength of the coupling between the phase and diffusion fields. The results of the asymptotic analysis imply that first the phase field simulations can be performed with a mesoscopic W, which tremendously decreases the computational time and opens up the possibility of performing simulations at smaller undercoolings; additionally, it is also possible to perform simulations with vanishing interface kinetics by choosing $\lambda = \tau D/W^2 a_2$. The limit of zero interface kinetics is physically relevant at low undercooling for a large class of materials.

In the case of unequal solid and liquid diffusion coefficients, Almgren [74] demonstrated that several non-equilibrium effects arise at the solid–liquid interface. These effects include a temperature jump across the interface, interface stretching and interface diffusion. In binary alloys, where the solute diffusivity in the liquid is essentially much larger than in the solid, these non-equilibrium effects translate to a discontinuity of the chemical potential of the solute element across the interface, solute diffusion along the arc length of the interface (surface diffusion) and the local increase in the arc length of a moving curved interface (interface stretching).

Karma [33] proposed a phase field model of a dilute binary alloy solidification. By performing a thin interface analysis, he demonstrated that the model reduces to a free-boundary problem with a modified mass conservation condition, which under the condition of vanishing solute diffusivity has the form

$$c_L(1-k_E)v_n = -D_L \partial_n c_L|_L + b_1 c_L(1-k_E) W \mathcal{K} v_n - b_2 W D_L \partial_S^2 c_L, \tag{46}$$

where c_L is the solute concentration on the liquid side of the interface, k_E is the equilibrium partition coefficient, D_L is the diffusion coefficient in the liquid, $\mathcal{K}$ is the local curvature of the interface and b_1, b_2 are coefficients of order unity. The partial derivatives in normal and tangential directions are denoted by ∂_n and ∂_S, respectively. The two last terms on the right-hand side are corrections corresponding to the stretching of the arc length of the interface due to the curvature and motion ($b_1 c_L(1-k_E) W \mathcal{K} v_n$) and due to solute diffusion along the interface ($b_2 W D_L \partial_S^2 c_L$). The discontinuity of the chemical potential corresponds to the departure from the chemical equilibrium at the interface and leads to solute trapping; this effect is highlighted in section 3.1.2. For diffuse interface models the solute trapping effect becomes important when the interface velocity approaches a characteristic value $V = D_L/W$. Therefore, all three non-equilibrium effects depend on the interface thickness. In real materials at low velocity solidification these effects are negligible. However, the use of mesoscopic diffuse interfaces artificially magnifies the non-equilibrium effects and significantly influences the evolution of pattern formation.

Karma proposed to eliminate this anomalous solute trapping by introducing the antitrapping current into the solute diffusion equation

$$\boldsymbol{j}_{AT} \sim c_L(1-k_E) W \frac{\nabla\phi}{|\nabla\phi|} \frac{\partial\phi}{\partial t}. \tag{47}$$

The antitrapping term produces a solute flux from the solid to the liquid along the direction normal to the interface $\vec{n} = -\nabla\phi/|\nabla\phi|$. The magnitude of this antitrapping current can be chosen to recover the local equilibrium at the solid–liquid interface. Along with the specially chosen interpolation functions, the antitrapping current gives the flexibility of eliminating all the non-equilibrium effects. Moreover, the model for binary alloy solidification [33] possesses the same computational benefits as the symmetrical model [31, 32].

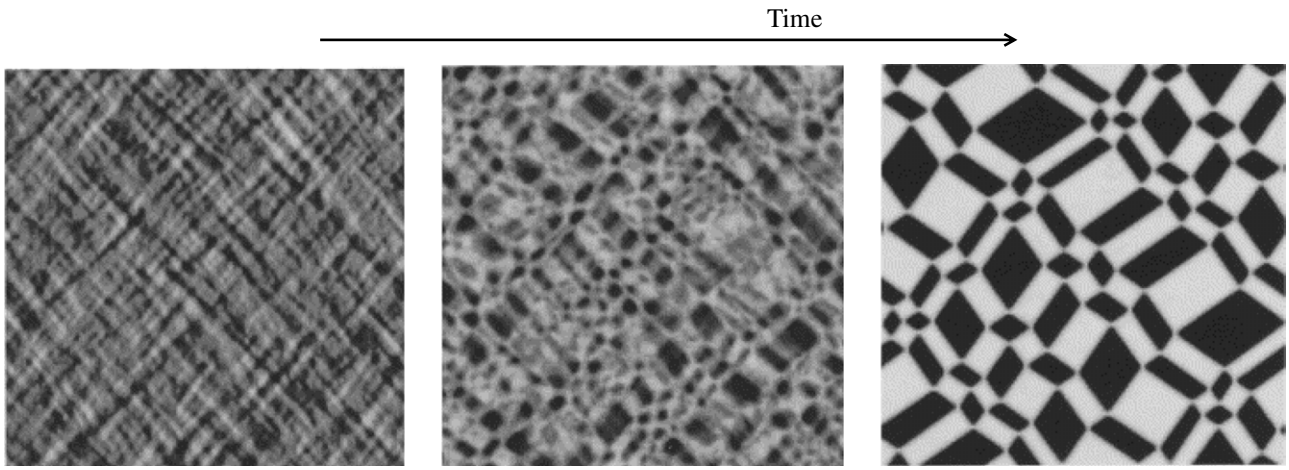


Figure 10. Temporal evolution of the microstructural evolution of a chess board pattern in solid–solid phase transformations (courtesy of Le Bouar *et al* [323]). Reprinted from [323], copyright (1998), with permission from Elsevier.

3.8. Solid-state transformations

Solid–solid phase transitions produce a rich variety of fascinating patterns, which have attracted the attention of many scientific groups. Phase field methods have been recognized as a reliable tool for modeling solid–solid phase transformations, see, e.g., detailed reviews by Ode [315] and by Chen [19]. In this paper we highlight areas of solid-state transitions where the phase field technique has been successfully applied.

In the modeling of solid–solid systems the phase field variable is often called an order parameter since it can be associated with the well-defined physical long-range order parameter for the order–disorder transformations. Essentially all solid–solid phase transformations result in the formation of coherent microstructures with a lattice mismatch at the boundary of neighboring particles. The degree of the lattice mismatch, the elastic properties of the material and the spatial distribution of solid particles define the magnitude of the elastic energy, which plays an important role in the microstructure evolution of coherent systems and has received considerable attention in phase field modeling.

A large number of phase field works are devoted to studying the kinetics of coarsening. These studies are based on 2D [200, 201, 316–319] and 3D [316, 320, 321] simulations in which many particles of an ordered precipitate phase evolve in a disordered matrix. With a sufficient number of particles, it is possible to monitor how the average particle size changes in time. It is demonstrated that the elastic energy has strong effects on the coarsening behavior [319, 322] and the microstructural development (see, e.g., figure 10 and for more details [323]).

Similar to phase field models of solidification, works towards quantitative modeling of coarsening processes [324] and comparison with the sharp interface model [325] have been reported. The phase field method has also been applied to investigate proper and improper martensitic transformations in single and polycrystals [326–331], cubic-to-tetragonal transformations [332–334], hexagonal-to-orthorhombic transformations [335–338], ferroelectric transformations [339, 340], phase transformations under applied stress [327, 330, 341–343] and in the presence of strain [344–346]. Attempts to describe the dynamics of nanovoids and nanobubbles in solids [347], particle splitting [348], dynamics of dislocations [349–352] and formation of dislocation networks [353, 354], development of zirconium hydride precipitation [355–357] and microstructural evolution in the presence of structural defects [352, 358] have been performed.

Nucleation has been incorporated into solid-state phase field models to simulate bimodal particle size distributions observed in continuous cooling [175, 318, 359]. A model describing viscoplastic behavior of eutectic tin solder alloys has been derived in [360, 361]. Results of atomistic simulations were used for constructing surface and elastic energies in the phase field model for precipitate microstructure evolution in binary alloys [362, 363].

Phase field models for precipitations in Ti–Al alloys were presented in [126, 364].

4. Numerical methods for solving the phase field equations

4.1. The finite difference method and time discretization

Most numerical work with the phase field model has been performed with the finite difference discretization on uniform structured grids, for example [14, 37, 38, 176]. This can be explained by the fact that the finite difference techniques are conceptually simple, and more importantly, the implementation of a finite difference algorithm is straightforward and very compact. Typically second order central differences are used, though the application of more sophisticated stencils have been reported [72, 365] in order to reduce the effects of the computational grid anisotropy. These effects were quantified by Mullis [366]. They are especially pronounced during dendritic simulations with low surface energy anisotropy.

Temporal discretization of the time derivatives can be done using explicit (e.g. forward Euler, Runge–Kutta) or implicit (e.g. backward Euler, Crank–Nicholson) methods. For the explicit schemes, the solution on the next time step is explicitly formulated using the current approximation. For implicit

schemes it is necessary to actually solve a large system of algebraic equations to update the solution. It is then not surprising that most of the studies based on the phase field method utilize explicit schemes.

The explicit time discretization works well when the fields considered in the problem have similar mobility/diffusivity rates (e.g. [36, 38]). This implies that similar demands are imposed on the stability of the numerical scheme. The stability condition relates the time step and the grid spacing, which must hold to prevent an uncontrolled growth of errors. However, when several diffusion fields are included into the model, their diffusivity may differ by several orders of magnitude, imposing unreachable demands on the time step. Consequently, implicit time discretization is preferred for complex systems (e.g. [39]).

The finite difference methods lead to structured systems approximating PDEs, which are easy to parallelize by the use of, e.g., MPI (message passing interface) [115, 176, 367]. George and Warren [115] provide a detailed description on code parallelization and visualization of the results in large 3D computational domains.

While solving phase field equations with an explicit finite difference discretization, one may reduce the computational time by the following trick proposed by Wang and Sekerka [63]: since the phase field variable differs from constant values only at the phase interface, one should solve the phase field equation only when values of the phase field variable entering the computational stencil differ.

4.2. Methods based on adaptive grids

Diffuse interface methods for modeling microstructural evolution feature large-scale separations. Commonly several orders of magnitude must be resolved: the thickness of the phase boundary is in the order of nanometres, while characteristic features of a microstructure are typically observed on a scale of micrometres or even millimetres. This scale separation is the main computational drawback of the phase field methods, often preventing the results from being quantitative. On the other hand, the variations of the phase field and diffusion fields are typically localized over the interfacial region. This motivates one to use adaptive grids, which would track the migration of the phase boundary.

Provatas *et al* [368] presented a finite element solution of dendritic growth on adaptive unstructured grids in 2D. In their adaptive code a dynamic quad-tree data structure was implemented. The grids consisted mostly of quadrilateral elements, which were combined with triangular elements to avoid hanging nodes. Tönhardt and Amberg [144–146] studied convective effects on dendritic growth using FEM on adaptive triangular grids. This code was also applied to simulate dendritic solidification of a binary alloy [39]. A finite element method on octagonal elements was used in a three-dimensional refinement algorithm developed by Jeong [80]. The code was parallelized and the simulations were performed on 16 processors. Burman *et al* [369] applied adaptive finite elements with high aspect ratios for simulations of coalescence.

Braun and Murray [62] performed calculations using a general adaptive finite difference technique coupled with a phase field model. Lan *et al* [153, 184] demonstrated the potential of the finite volume method for simulating dendritic growth in two dimensions on adaptive quadrilateral grids. They used domains as big as $200\,000 \times 200\,000$ nodes for simulating pure and binary dendritic growth with flow. The benefit of applying adaptive finite volume methods is that hanging nodes do not present a problem. Also these rather structured grids are potentially easy to partition for calculations in a parallel environment. The parallelization of adaptive finite element codes in two and three dimensions was performed by Do-Quang *et al* [370]. In the code, dynamic mesh partitioning for load balancing was performed after every grid change. By means of this code, 2D simulations of Widmanstätten plates [50] and a part of the simulations reported in [91] were performed. Opposite to the finite difference method, finite element and finite volume approaches require careful programming, which may by itself take a lot of time. However, this problem can be overcome, e.g., by automatic code generation, as implemented in the symbolic computational package femLego [371, 372].

4.3. Other approaches

A hybrid finite-difference-diffusion-Monte-Carlo method for diffusion-limited growth problems was proposed by Plapp and Karma [373]. Their idea is based on a multiscale diffusion Monte Carlo algorithm, which allows off-lattice random walkers to take longer and computationally rare steps with increasing distance away from the phase boundary. The computational domain is split into two parts: the first one contains the solid phase and a thin layer of the liquid phase surrounding the interface, the second part contains the remaining liquid. The phase field equations are solved by means of finite differences in the first domain, while the large-scale diffusion field is represented by an ensemble of off-lattice random walkers and is evolved using the diffusion Monte Carlo method. The two solutions are connected at some distance from the moving interface. The method has been applied to simulate dendritic growth of a pure substance [70, 72]. The performance of the method is similar to the performance of the adaptive finite elements approach.

There have been many attempts to derive efficient numerical techniques for solving phase field systems. Multigrid methods show good performance for simulating dendritic growth with fluid flow [151] and directional solidification [119]. Another approach common in directional solidification is to use a moving computational domain, which is adjusted to the velocity of the solidification front (see, e.g., [180]). A moving mesh method is a simple and efficient way to solve the equations in 1D [374]. Chan and Shen [375] proposed to use a semi-implicit Fourier-spectral method to calculate the evolution of systems dominated by long-range elastic interactions. The wavelet-Galerkin scheme developed by Wang and Pan [376] appears to be a powerful computational tool for solving the phase field equations. Badalassi *et al* [377] presented a novel time-split method for solving the coupled Cahn–Hilliard and Navier–Stokes equations and demonstrated the performance of the method for the phase separation in two

and three dimensions. The use of lattice-Boltzmann methods to solve the phase field equations coupled to fluid flow was shown to work efficiently [159, 162, 378]. The numerical integration scheme of Vollmayr-Lee and Rutenberg [379] is unconditionally stable, allows for variable time steps and reveals a remarkable computational speedup compared with the Euler method.

4.4. Computational performance of the numerical methods

Before applying a numerical scheme for solving the phase field equations, one must choose the one which fits best to the expected behavior of the system. For dendritic growth, numerical experiments demonstrate that finite difference codes are most efficient in the case of high driving forces, when the diffusion length is small and the dendritic microstructures are highly branched. The application of finite element and finite volume methods on adaptive grids is essentially effective in the low undercooling/supersaturation limit where the dendrite tip radius is one or more orders of magnitude smaller than the characteristic spatial scale of variation of the surrounding diffusion fields. In both cases, the parallelization of the computational codes significantly speeds up the simulations and allows for the use of large computational domains.

4.5. Finite difference discretization of the anisotropic Laplacian operator

Probably, the most difficult part in solving the phase field equations by the finite difference method is the correct discretization of the anisotropic Laplacian operator. Here we present a second order accurate discretization on a uniform cubic grid, whose nodes have coordinates $x_i = i\Delta x$, $y_j = j\Delta y$, $z_k = k\Delta z$ and Δx, Δy and Δz are the grid spacing in x, y and z directions. Before presenting the numerical scheme, let us notice that the operator (40) with η given by (41) is a sum of the following expressions:

$$\begin{aligned}
&\frac{\partial}{\partial\xi}\left(\eta^2(\vec{n})\frac{\partial\phi}{\partial\xi}\right)+\frac{\partial}{\partial\xi}\left(|\nabla\phi|^2\eta(\vec{n})\frac{\partial\eta(\vec{n})}{\partial\phi'_\xi}\right)\\
&\quad=\frac{\partial}{\partial\xi}\left[\left\{\eta^2(\vec{n})+\eta(\vec{n})\right.\right.\\
&\qquad\left.\left.\times 16\gamma\left(\frac{(\phi'_x)^2}{|\nabla\phi|^2}-\frac{(\phi'_x)^4+(\phi'_y)^4+(\phi'_z)^4}{|\nabla\phi|^4}\right)\right\}\frac{\partial\phi}{\partial\xi}\right]\\
&\quad=\frac{\partial}{\partial\xi}\left(f(\phi'_x,\phi'_y,\phi'_z)\frac{\partial\phi}{\partial\xi}\right), \qquad (48)
\end{aligned}$$

where ξ corresponds to x, y, z and f is a function of the derivatives ϕ'_x, ϕ'_y and ϕ'_z. To discretize the last expression at the grid node x_i, y_j, z_k we apply the standard central differences

$$\begin{aligned}
\left.\frac{\partial}{\partial x}\left(f\frac{\partial\phi}{\partial x}\right)\right|_{i,j,k}&=\frac{1}{2\Delta x^2}\left[\left(f_{i+1,j,k}+f_{i,j,k}\right)\left(\phi_{i+1,j,k}-\phi_{i,j,k}\right)\right.\\
&\left.-\left(f_{i,j,k}+f_{i-1,j,k}\right)\left(\phi_{i,j,k}-\phi_{i-1,j,k}\right)\right] \qquad (49)
\end{aligned}$$

and the first derivatives in f are approximated by

$$\phi'_x=\frac{\phi_{i+1,j,k}-\phi_{i-1,j,k}}{2\Delta x}. \qquad (50)$$

Discretization of the y and z derivatives is performed in the same way.

5. Concluding remarks, outstanding problems

In this paper we have reviewed recent advances of the phase field modeling technique, a method which proved to be beneficial for the numerical simulation of phase transformations. While the method of phase fields was proposed already by Langer [3] in the 1980s only the recent increase in computational power allowed accurate calculations with realistic interface thicknesses and undercoolings. Phase field models have been proposed for a multitude of phenomena starting from pure substances growing into undercooled melt and binary alloy solidification up to multicomponent systems. Realistic modeling of phase transformations evidently also includes the influence of natural convection and noise. Phase field models for different settings such as free growth, directional solidification, grain growth and multiphase transformations as well as solid–solid transformations have been discussed.

The phase field modeling technique was thoroughly developed over the years and has now become a powerful and reliable method in computational physics and material science. However, much still remains to be done. The poor agreement with solvability theory for high anisotropies opens some questions about the validity range of either the phase field models or the solvability theory. Whether an improved theory has to be developed or correction terms in the phase field equations introduced is an open question.

While more efficient algorithms such as adaptive grids were able to decrease the problem of multiscale behavior of solidifying systems up to a certain extent, solidification for low undercoolings as found in experiments is still too difficult to solve due to the separation of scales. The recent semi-sharp approach of Amberg [266] might remedy this situation; this however still remains to be proved in practical applications. For binary alloy solidification it seems that the antitrapping current introduced by Karma [33] eliminates the anomalous solute trapping effects and correct tip radii and tip velocities are found; however, the concentration profiles observed for small interface thicknesses without an antitrapping term do not match the ones for larger interface thickness and antitrapping included. While the inclusion of realistic noise in the equations using a Langevin term for nucleation was shown to work, it is not yet possible to add this kind of noise without great loss of computational efficiency. Fast generation of Langevin noise is thus needed. A method for simulating real, non-dilute N-component systems has not been developed so far, and simulations were only performed under restrictive simplifications such as dilute alloys and neglecting off-diagonal components of the diffusion matrix. For grain growth, as well as for polycrystalline systems, an elegant method of simulating N arbitrary crystals without having to resort to N phase fields is still lacking. Although promising progress has been made by Warren and Kobayashi [223] the problem is still essentially unsolved. On the physical side an understanding of the influence of the shape and distribution

of initial nuclei on the resulting microstructures as well as a deeper insight into transient stages of the growth process is highly desirable.

New approaches such as the PFC description on atomistic length scales and time scales relevant for pattern formation processes show that the general framework of deriving the dynamics of systems governed by free energy functionals is far from being exhausted. While the PFC model is able to bridge the mesoscopic phase field model to the atomistic view, the way the energy functional depends on the non-linear lattice terms is not transparent. In particular, a method to derive these non-linear terms for a given lattice structure is missing.

The phase field modeling technique, although first developed for phase transformations, seems to be an interesting method for other fields, such as in biology or electrochemistry, where the approach is also suited to describing vesicle dynamics, epitaxy and monolayer evolution.

Future research surely will include different directions: while the phase field method has definitely reached a certain maturity, essential methods for industrial use are still lacking (in particular, N-component systems and polycrystalline systems); hence, considerable efforts will be made for industrial research. The trend to use phase field models as a method of choice for modeling phase transformation will surely continue. Further investigation on atomistic scales and eventually the connection of phase field models and PFC models might prove to be fruitful; in particular, bridging length scales from atomistic to macroscale seems to be a conceivable, albeit daunting task. Additionally the phase field model will continue to expand to different research fields, where moving boundary problems also occur.

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